UPPER GASTROINTESTINAL ONCOLOGY

Gastric & Esophageal Cancer Management 2026

VISUAL CLINICAL ATLAS
3D ILLUSTRATED EDITION

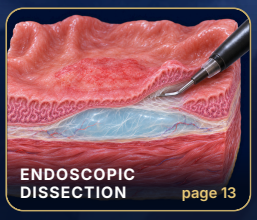
From Diagnosis & Staging to Survivorship

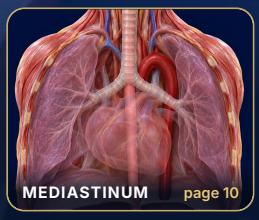
NCCN v4.2026 ×2 | ESMO Guideline · Oct 2026 | ASCO Living Guideline v2026.1.3 | ASCO · ESMO 2026 congress data

NEW **13 high-definition 3D plates inside** Anatomy · endoscopic dissection · every operation · D2 map | click a picture to open its page















NON-NEGOTIABLE
Red-flag steps that must not be skipped in any patient

NEGOTIABLE
Where the MDT individualizes: options with equipoise

ASCO 2026
Guideline update and congress data, color-coded in blue

ESMO 2026
New guideline and Madrid congress data, color-coded in green

12
chapters, each a decision algorithm plus an infographic

13
3D plates in the anatomy and surgical atlases

108
trials in a six-page compendium with FDA approval dates

43
pages, built for the tumor board and the clinic

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READER'S GUIDE

How to read this atlas

ESOPHAGUS STOMACH

One atlas for both upper-gastrointestinal cancers, because the 2026 guidelines treat them by **histology** rather than by organ: squamous cell carcinoma of the esophagus, and adenocarcinoma of the esophagus, junction and stomach.

First page · Decision algorithm
From the clinical question to the treatment, then the red-flag and negotiable boxes for that step.

Second page · Infographic
The numbers and trials behind the pathway, with the ASCO (blue) and ESMO (green) updates of 2026.

3D Plates · chapters 3, 4 and 8
Thirteen three-dimensional models of the anatomy, endoscopic dissection and every operation, each with a numbered key.

COLOR KEY

NON-NEGOTIABLE
A **red flag**: a step that must not be skipped or traded away in any patient — it protects cure, safety or the validity of every later decision.

NEGOTIABLE
Genuine equipoise: more than one defensible option. **Individualize at the MDT** by fitness, biology, resources and patient preference.

ASCO 2026 Blue
ASCO guideline update and living guideline, ASCO GI (Jan) and the ASCO Annual Meeting (May–Jun 2026).

ESMO 2026 Green
The new ESMO Clinical Practice Guideline (Oct 2026), ESMO GI (Jul) and the ESMO Congress, Madrid (Oct 2026).

Inline tags you will meet

ASCO 2026 ESMO 2026 source of an update
FDA Aug 2026 US approval (month, year)
Regional data or approval limited to one region
Cat 1 I, A NCCN category / ESMO level, grade
Negative trial missed its primary endpoint
Red flag Negotiable as in the boxes above

BUILT ON

Source	Version used in this edition	What it anchors
NCCN Guidelines®	Esophageal & EGJ Cancers **v4.2026** (16 Sep 2026) · Gastric Cancer **v4.2026** · Genetic/Familial High-Risk: Colorectal, Endometrial, Esophageal & Gastric **v1.2026** (16 Jun 2026)	Algorithms, categories of evidence, surgery and radiotherapy principles
ESMO	**Gastric and oesophageal cancer: ESMO Clinical Practice Guideline** — Ann Oncol, online 15 Oct 2026 (first merged guideline, organized by histology)	Levels of evidence, ESMO-MCBS, European label thresholds
ASCO	**Immunotherapy & Targeted Therapy for Advanced Gastroesophageal Cancer**: Guideline Update (J Clin Oncol 2026;44:1145) and Living Guideline **v2026.1.3** (22 Sep 2026)	Biomarker-driven first- and later-line therapy
Staging	AJCC Cancer Staging Manual, **8th edition** (still the current system for esophagus/EGJ and stomach in 2026) · JGCA Treatment Guidelines, 6th edition	TNM, stage groups, endoscopic-resection and D2 definitions
Congress data	ASCO GI (8–10 Jan) · ASCO (29 May–2 Jun) · ESMO GI, Munich (1–4 Jul) · ESMO, Madrid (23–27 Oct 2026)	Color-coded update boxes and the trials compendium
Regulatory	FDA approvals to November 2026; EMA noted where the label differs	Approval dates in the compendium

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Abbreviations. AC adenocarcinoma · CPS combined positive score · CRT chemoradiotherapy (nCRT neoadjuvant, dCRT definitive) · D-FLOT durvalumab + FLOT · dMMR mismatch-repair deficient · EFS event-free survival · EGJ/GEJ esophagogastric junction · ER endoscopic resection · ESCC esophageal squamous cell carcinoma · ESD endoscopic submucosal dissection · FP fluoropyrimidine · GEA gastroesophageal adenocarcinoma · HR hazard ratio · ICI immune checkpoint inhibitor · MDT multidisciplinary team · MSI-H microsatellite instability-high · OS overall survival · pCR pathological complete response · PFS progression-free survival · TAP tumor area positivity · TPS tumor proportion score · T-DXd trastuzumab deruxtecan.
Scope. An educational summary for oncology professionals; it does not replace the guidelines, the product labels or the treating team's judgment. Check doses against the current label and local protocol. 3D plates: computer-generated teaching models, not patient images. Evidence current to 13 November 2026.

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A

ASCO 2026 UPDATE DIGEST · COLOR-CODED BLUE THROUGHOUT THE ATLAS

ASCO 2026: what changed

ESOPHAGUS STOMACH

≥1 PD-L1 entry threshold for first-line immunotherapy, both histologies

4 biomarkers before first-line therapy: PD-L1, MMR/MSI, HER2, CLDN18.2

26.4 mo median OS, zanidatamab + tislelizumab + chemo (HERIZON-GEA-01)

v1.3 living guideline v2026.1.3, 22 Sep 2026: revised as evidence arrives

ASCO GUIDELINE — Immunotherapy and targeted therapy for advanced gastroesophageal cancer: Guideline Update (Feb 2026) and Living Guideline

Biomarker profile (advanced disease)	First-line recommendation	What moved since 2023
Test first. Adenocarcinoma: PD-L1, MMR/MSI, HER2, CLDN18.2. ESCC: PD-L1, MMR/MSI. Consider broad NGS.	Results in hand **before** the first-line choice	New good-practice statement; CLDN18.2 is new
HER2−, pMMR · PD-L1 ≥1 · CLDN18.2−	Chemotherapy doublet **+ immunotherapy** (nivolumab, pembrolizumab or tislelizumab)	One PD-L1 ≥1 entry point; benefit grows with expression
HER2−, pMMR · PD-L1 <1 · CLDN18.2+	Chemotherapy **+ zolbetuximab**	New recommendation
HER2−, pMMR · PD-L1 ≥1 *and* CLDN18.2+	Immunotherapy **or** zolbetuximab with chemotherapy **Negotiable**	New; no head-to-head data
HER2−, pMMR · PD-L1 <1 · CLDN18.2−	Chemotherapy doublet alone	Immunotherapy not recommended
HER2-positive, pMMR	**HER2-targeted antibody** + doublet chemotherapy; **add immunotherapy if PD-L1 ≥1**	Living update v2026.1.2 (Jul 2026), after HERIZON-GEA-01
dMMR / MSI-H	Immunotherapy + fluoropyrimidine/oxaliplatin; immunotherapy alone case by case	Stand-alone recommendation
ESCC, pMMR	PD-L1 ≥1: immunotherapy + chemotherapy, **or** nivolumab + ipilimumab. PD-L1 <1: chemotherapy alone	One threshold replaces CPS ≥10 / TPS ≥1%
Second line and beyond	Ramucirumab + paclitaxel · **T-DXd** if HER2+ · ESCC, no prior ICI: nivolumab or tislelizumab (PD-L1 ≥1), pembrolizumab (CPS ≥10)	ESCC second line added

Shah MA, Kennedy EB, Deighton D, et al. J Clin Oncol 2026;44:1145–1165; ASCO Living Guideline v2026.1.3, 22 Sep 2026.

ASCO GI 2026 — HERIZON-GEA-01 primary analysis (LBA285): first-line HER2-positive adenocarcinoma, n = 914

Zanidatamab, a bispecific HER2 antibody, + chemotherapy ± tislelizumab against trastuzumab + chemotherapy. **PFS 12.4 vs 8.1 months** in both zanidatamab arms: HR 0.63 with tislelizumab, 0.65 without (N Engl J Med 2026;394:2002).

Zanidatamab + tislelizumab + chemo
HR 0.72 (0.57–0.90), p = 0.004



Trastuzumab + chemo
control arm



FDA 25 Aug 2026 with tislelizumab: IHC 3+ or IHC 2+/ISH+, any PD-L1. Without: IHC 3+ only.

IN REVISION — Locally advanced esophageal carcinoma

- The 2020 guideline is being **updated in the 2025–26 cycle.**
- The 2020 text predates **ESOPEC, MATTERHORN** and the final CheckMate 577 survival data.
- Until then: NCCN v4.2026 and ESMO 2026 (chapters 5–6).

ASCO 2026 — Annual Meeting, Chicago, 29 May – 2 June 2026

ASCO Annual Meeting 2026	Design	Result	Practice signal
PANKU-Esophagus01 izalontamab brengitecan, EGFR×HER3 ADC	Phase 3, n = 497, China. ESCC after platinum + PD-(L)1; vs irinotecan or a taxane	OS **9.8 vs 7.2 mo**, HR 0.63 (0.48–0.82); PFS 4.2 vs 2.0 mo, HR 0.50 (JAMA 2026)	Regional First ADC to beat chemotherapy after immunotherapy; approved in China only
ATTRACTION-6 nivolumab + ipilimumab + chemo	Phase 3, n = 626, East Asia. First-line HER2− gastric/GEJ; vs chemotherapy	OS 15.7 vs 15.8 mo, HR 0.90 (0.74–1.09); more toxicity	Negative Dual checkpoint blockade + chemotherapy is **not** a standard
HERIZON-GEA-01 by PD-L1	Pre-planned subgroups	Benefit of the tislelizumab triplet in PD-L1-positive and -negative tumors	Supports the PD-L1-agnostic FDA label
SCIENCE neoadjuvant sintilimab + chemo or CRT	Phase 3, China. Resectable ESCC; vs CRT	Higher pCR with immuno-chemoradiotherapy; EFS immature	Regional Watch: no survival data yet

ASCO 2025 — Reported a year earlier

ASCO 2025, decisive for 2026	Result	Consequence
CheckMate 577 final OS	51.7 vs 35.3 mo, HR 0.85 (0.70–1.04), p = 0.106; 5-year OS 46% vs 41%; DFS HR 0.76	Adjuvant nivolumab stays for residual disease after CRT + surgery; benefit concentrated in PD-L1 CPS ≥1
DESTINY-Gastric04	T-DXd vs ramucirumab + paclitaxel: OS 14.7 vs 11.4 mo, HR 0.70 (0.55–0.90)	T-DXd is the second-line standard in confirmed HER2+ disease
MATTERHORN EFS	2-year EFS 67.4% vs 58.5%, HR 0.71; pCR 19.2% vs 7.2%	Perioperative durvalumab + FLOT **FDA Nov 2025**

ASCO IN ONE LINE | ASCO in 2026: **test four biomarkers first**, give immunotherapy only when **PD-L1 ≥1**, bring **zolbetuximab** and **zanidatamab** into the first line, and drop dual checkpoint blockade with chemotherapy.

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ESMO 2026: what changed

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ESMO GUIDELINE Gastric and oesophageal cancer: ESMO Clinical Practice Guideline. Annals of Oncology, online 15 October 2026

- **One guideline, two histologies:** the 2022 guidelines are merged and organized as squamous cell carcinoma versus gastroesophageal adenocarcinoma.
- **Four biomarkers at diagnosis** of adenocarcinoma: HER2 (I, A), PD-L1 by CPS or TAP (I, A), CLDN18.2 (I, A), MSI/MMR (II, A). Squamous: PD-L1. **DPD testing** before any fluoropyrimidine.
- **Durvalumab + FLOT** preferred for resectable adenocarcinoma, ≥cT2 or cN+, where available (I, A) (MCBS A).
- **FLOT before CROSS** for esophageal and junctional adenocarcinoma; chemoradiotherapy if unfit for FLOT, then adjuvant nivolumab when PD-L1-positive (I, B).
- **Squamous:** CROSS, restaging and surgery (I, A); surgery still recommended after a clinical complete response; adjuvant nivolumab for residual disease (I, A).
- **Definitive chemoradiotherapy, 50 Gy in 25 fractions,** for cervical tumors and patients unfit for or declining surgery (III, B).
- **HER2+, first line:** zanidatamab + tislelizumab + chemotherapy where available (I, A); zanidatamab + chemotherapy for IHC 3+ (I, B); otherwise trastuzumab + chemotherapy, plus pembrolizumab if CPS ≥1.
- **Trial-specific PD-L1 cut-offs kept:** nivolumab CPS ≥5, pembrolizumab CPS ≥1, tislelizumab TAP ≥5%.
- **T-DXd** in second or third line after trastuzumab (I, A); **re-biopsy** first: about 30% lose HER2 positivity.
- **No surgery for metastatic disease** (REGATTA, RENAISSANCE). Follow-up 3-monthly for 2 years, then 6-monthly to 5 years (IV, B).

Al-Batran S-E, Bencivenga M, Crosby T, et al., on behalf of the ESMO Guidelines Committee. Ann Oncol 2026; doi 10.1016/j.annonc.2026.09.015. Level of evidence and grade in brackets.

ESMO 2026 ESMO Congress, Madrid, 23–27 October 2026

ESMO Congress 2026 · Madrid	Design	Result	Practice signal
KEYNOTE-975 (2875O) pembrolizumab + definitive CRT	Phase 3. Unresectable esophageal/GEJ cancer; vs placebo + CRT	EFS 15.2 vs 14.7 mo, HR 0.97 (0.77–1.23), CPS ≥10; OS HR 1.00; stopped for AEs 34% vs 27%	Negative Do **not** add pembrolizumab to definitive chemoradiotherapy
HERIZON-GEA-01 (LBA79) updated OS at 35 months	Phase 3, n = 914. First-line HER2+	Zanidatamab + chemo **22.9 vs 19.6 mo**, HR 0.76 (0.62–0.93); with tislelizumab **27.0 mo**, HR 0.65 (0.52–0.80); triplet vs doublet HR 0.81, p = 0.050	OS gain now significant for the doublet; extra value of tislelizumab not formally proven
G-HOPE-001 (LBA77) arcotatug tavatecan, CLDN18.2 ADC	Phase 3, China + Japan. CLDN18.2+, HER2−, 2–4 prior regimens; vs chemotherapy	OS **10.2 vs 6.9 mo**, HR 0.57 (0.45–0.73); PFS 7.1 vs 3.1 mo; ORR 37% vs 8%	Both primary endpoints met; awaits regulatory decisions
CLARITY-Gastric01 (LBA7) sonesitatug vedotin, CLDN18.2 ADC	Phase 3. CLDN18.2 ≥25%, pretreated; vs investigator's choice	Third line+: OS **10.0 vs 5.0 mo**, HR 0.47 (0.36–0.62); PFS co-primary missed (p = 0.053)	Second CLDN18.2 ADC with an OS gain; not yet approved
KC-WISE (LBA80) anbenitamab, HER2 bispecific, + chemo	Phase 3, n = 246, China. HER2+ after trastuzumab	OS **19.6 vs 9.6 mo**, HR 0.39 (0.28–0.54); grade ≥3 events 69% vs 48%	Regional Strong signal, marked hematologic toxicity
LOMAC (LBA81) 4 vs 6 months of adjuvant XELOX	Phase 3 non-inferiority, n = 1,023. Stage II–IIIB after D2	3-year DFS 69.2% vs 70.8%, HR 1.01 (0.82–1.25); grade ≥3 neuropathy 0.4% vs 5.9%	Negotiable Four months is a valid option after D2
RAMIRIS (2878RO) FOLFIRI + ramucirumab	Phase 3. Second line after a taxane; vs paclitaxel + ramucirumab	OS 8.7 vs 7.5 mo, HR 0.95; PFS 5.5 vs 3.7 mo, HR 0.71	Negative for OS; option after a taxane
DEWI (phase 2) dostarlimab, then watch and wait	Localized dMMR/MSI-H adenocarcinoma	15 of 24 (62.5%) in clinical complete response without surgery at 1 year	Negotiable Organ preservation: investigational

ESMO GI · ESMO 2025 The rest of the ESMO evidence behind 2026 practice

Also in the ESMO calendar	Result	Consequence
STAR-221 · ESMO GI, Munich, Jul 2026	Domvanalimab (anti-TIGIT) + zimberelimab + chemotherapy: no OS gain over nivolumab + chemotherapy	Negative No role for anti-TIGIT in the first line
MATTERHORN final OS · ESMO 2025	HR 0.78 (0.63–0.96), p = 0.021; 3-year OS 69% vs 62% (Lancet 2026)	FDA Nov 2025 and EC approval (Mar 2026)
SKYSCRAPER-07 · ESMO 2025; Ann Oncol 2026	After definitive CRT (n = 760), tiragolumab + atezolizumab missed its endpoint; atezolizumab alone: OS HR 0.69 (0.52–0.91), descriptive	Hypothesis-generating; not licensed
FORTITUDE-101 · ESMO 2025	Bemarituzumab + mFOLFOX6, FGFR2b ≥10%: OS 17.9 vs 12.5 mo (HR 0.61), fading to 14.5 vs 13.2 (HR 0.82); corneal toxicity	Not practice-changing
PERISCOPE II · ESMO 2025; Lancet Oncol 2026	Gastrectomy + cytoreduction + HIPEC vs systemic therapy (n = 102): OS 15.7 vs 16.6 mo, HR 1.10; serious adverse events 44% vs 6%	Negative No HIPEC outside trials

Same evidence, different reading (PD-L1 thresholds: chapter 2)	ESMO 2026	ASCO 2026 · NCCN v4.2026 · FDA
Clinical complete response after CRT, squamous	Surgery still recommended	NCCN: surveillance is an option in selected patients
Durvalumab + FLOT when PD-L1 <1	Preferred where available, any PD-L1	NCCN category 2A (upgraded Sep 2026); category 1 when CPS/TAP ≥1

ESMO IN ONE LINE | ESMO in 2026: **durvalumab + FLOT** for resectable adenocarcinoma, **zanidatamab** for HER2+, CLDN18.2 testing for all, and three clear "no"s: pembrolizumab with definitive CRT, HIPEC, and surgery for metastatic disease.

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DECISION ALGORITHM

1 Epidemiology, risk & prevention

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An adult with risk factors or upper-gastrointestinal symptoms
Risk is histology-specific: read the three lanes separately

Squamous-cell risk

Who

- Tobacco **and** alcohol
- Previous head-and-neck or lung squamous cancer
- Achalasia, caustic stricture, tylosis
- Residence in a high-incidence belt (East Asia, East and southern Africa, Iran)

Endoscopic screening
High-definition endoscopy with **Lugol chromoendoscopy** or narrow-band imaging; organized programs only where incidence is high

Squamous dysplasia
Low grade: surveillance. High grade: **endoscopic resection** (chapter 4)

Adenocarcinoma risk (esophagus, junction)

Who

- Chronic reflux for 5 years or more **plus** risk factors: male sex, age over 50, central obesity, smoking, first-degree relative
- Known Barrett esophagus

Screen once for Barrett esophagus
Endoscopy, or a swallowable cell-collection device where available

Barrett surveillance

- No dysplasia: endoscopy every 3–5 years by segment length
- Confirmed low-grade dysplasia: endoscopic eradication, or surveillance every 6–12 months
- High-grade dysplasia or T1a: **endoscopic eradication** (chapter 4)

Gastric risk

Who

- *Helicobacter pylori* infection
- First-degree relative with gastric cancer; origin from a high-incidence region
- Atrophic gastritis, intestinal metaplasia, pernicious anemia, previous partial gastrectomy

Test and treat *H. pylori*
Eradicate, then **confirm eradication** at least 4 weeks after therapy

Endoscopic screening and surveillance

- National programs in high-incidence countries (every 2 years from 40 in Korea, from 50 in Japan)
- Extensive atrophy or intestinal metaplasia (OLGA/OLGIM III–IV): every 3 years

Any alarm symptom → upper endoscopy within 2 weeks
Dysphagia or odynophagia · unintentional weight loss · persistent vomiting · gastrointestinal bleeding or iron-deficiency anemia · early satiety · epigastric mass · new dyspepsia from the age of 55–60

Primary prevention: what is worth saying in clinic

Squamous
Stop tobacco and alcohol together; let very hot drinks cool; treat achalasia.

Junctional adenocarcinoma
Weight control and reflux treatment; stop smoking; keep Barrett surveillance.

Gastric
Eradicate *H. pylori* in patient and household; less salt-preserved food; stop smoking.

Hereditary risk: referral criteria and action

Who goes to genetics	Gene	What to do when a pathogenic variant is found
Every gastric cancer: tumor MMR/MSI; germline multigene panel	Lynch genes, *CDH1, CTNNA1, APC, TP53, STK11, SMAD4, BMPR1A, BRCA1/2, PALB2, ATM*	Syndrome-specific surveillance; cascade testing of relatives (NCCN Genetic/Familial v1.2026)
Diffuse gastric cancer before 50; two or more gastric cancers in a family, one diffuse; diffuse gastric and lobular breast cancer	*CDH1, CTNNA1*	**Risk-reducing total gastrectomy** in early adulthood, or yearly expert endoscopic surveillance if deferred; breast MRI from 30 in women
Gastric cancer with polyposis, or hamartomatous polyps	*APC* (FAP, GAPPS), *STK11, SMAD4, BMPR1A*	Upper endoscopy schedule by syndrome; gastrectomy for dysplasia or uncontrollable polyposis

NON-NEGOTIABLE — RED FLAGS

- **Alarm symptoms mean endoscopy now**, not a trial of acid suppression.
- ***H. pylori*: treat and prove eradication**; offer testing to first-degree relatives.
- **MMR/MSI on every gastric cancer**, and genetic referral when criteria are met.
- ***CDH1* carriers** are managed in an expert center: random-biopsy endoscopy misses most signet-ring foci.
- **Squamous cancer:** examine the head, neck and lungs for a second primary (about 7%).

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Population endoscopic screening in low-incidence countries.
- The threshold for Barrett screening, and non-endoscopic devices.
- Surveillance interval for intestinal metaplasia outside extensive disease.
- Timing of risk-reducing gastrectomy versus surveillance in *CDH1* carriers.
- Chemoprevention: high-dose proton-pump inhibitor with aspirin in Barrett esophagus.

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1

INFOGRAPHIC

Epidemiology, risk & prevention

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969,000
new **stomach** cancers a year worldwide
GLOBOCAN 2022 · 5th most common cancer

660,000
deaths from stomach cancer
5th cause of cancer death

511,000
new **esophageal** cancers a year
about 75% of them in Asia

445,000
deaths from esophageal cancer
7th cause of cancer death

Two diseases in one tube

	Squamous cell carcinoma	Adenocarcinoma of the esophagus and junction
Share of esophageal cancer	About 85% worldwide	About 14%; the majority in North America and northern Europe
Site	Upper and middle thirds	Lower third and junction, on Barrett esophagus
Drivers	Tobacco, alcohol, hot beverages, poor diet, betel quid	Reflux, obesity, smoking; male sex
Precursor	Squamous dysplasia	Barrett metaplasia → dysplasia
Biology	Resembles head-and-neck squamous cancer	Resembles chromosomally unstable gastric cancer
Curative backbone (chapters 5–6)	Chemoradiotherapy ± surgery	Perioperative chemotherapy + surgery

Helicobacter pylori

- A group 1 carcinogen: about **9 in 10 non-cardia gastric cancers** are attributable to it.
- Eradication roughly **halves** gastric-cancer incidence in randomized trials (relative risk about 0.54).
- Non-cardia disease is 82% of the global burden; cardia 18%.

Gastric adenocarcinoma: four molecular families (TCGA)

Chromosomal instability — intestinal type, junction and cardia; *TP53*; amplified *HER2, EGFR, MET, FGFR2* — **50%**

Microsatellite-unstable — hypermutated, *MLH1* silencing; older patients, antrum; immunotherapy-sensitive — **22%**

Genomically stable — diffuse histology; *CDH1, RHOA, CLDN18–ARHGAP* fusions — **20%**

EBV-positive — fundus and body; *PIK3CA*; PD-L1/PD-L2 amplification — **9%**

Risk factors at a glance

Squamous cell carcinoma
- Tobacco, alcohol (synergistic)
- Very hot drinks, betel quid
- Low fruit and vegetable intake
- Achalasia, caustic injury, tylosis
- Previous aerodigestive squamous cancer

Esophageal and junctional adenocarcinoma
- Gastro-esophageal reflux
- Barrett esophagus
- Central obesity
- Smoking
- Male sex, older age

Gastric adenocarcinoma
- *H. pylori*; Epstein–Barr virus
- Salt-preserved and smoked foods, smoking
- Atrophic gastritis, intestinal metaplasia
- Family history, hereditary syndromes
- Previous gastric surgery, pernicious anemia

Hereditary syndromes that raise gastric-cancer risk

Syndrome	Gene	Gastric-cancer risk	Key action
Hereditary diffuse gastric cancer	*CDH1* (*CTNNA1*)	About 30–40% by 80 in families meeting criteria; lower in unselected carriers	Risk-reducing total gastrectomy or expert surveillance
Lynch syndrome	*MLH1, MSH2, MSH6, EPCAM*	Up to about 10%, mainly *MLH1/MSH2*	Upper endoscopy with colonoscopy; treat *H. pylori*
FAP, GAPPS	*APC*	Fundic-gland polyposis; risk rises with dysplasia	Endoscopic surveillance of stomach and duodenum
Peutz–Jeghers	*STK11*	About 29%	Upper endoscopy from the late teens
Juvenile polyposis	*SMAD4, BMPR1A*	Up to about 21% with gastric polyps	Upper endoscopy from adolescence
Li–Fraumeni	*TP53*	Increased, earlier onset	Upper endoscopy within whole-body surveillance

Squamous cancer travels with second primaries

- **Head-and-neck second primary in 6.7%** (meta-analysis of 12 studies, n = 6,483): examine this region and the lungs at staging.

New in 2026 · NCCN Genetic/Familial v1.2026

- **Germline panel for every gastric cancer**, whatever the age (category 2B if diagnosed after 50).
- Esophageal cancer enters the hereditary-risk guideline for the first time.

KEY MESSAGE | Risk is written in the histology: **tobacco and alcohol** for squamous cancer, **reflux and obesity** for junctional adenocarcinoma, ***H. pylori*** **and heredity** for gastric cancer. Prevention starts with eradication and with the family history.

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DECISION ALGORITHM

2 Diagnosis, work-up & biomarkers

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Suspected esophageal or gastric cancer
Symptoms, an abnormal image, or a screen-detected lesion

Upper endoscopy with biopsies
6–10 biopsies of an advanced tumor (1–2 of an early lesion, then resect). Record: distance from the incisors, length, circumference, relation to the junction (Siewert type), Barrett segment; in the stomach, site and proximal–distal extent

Pathology
Squamous cell carcinoma or adenocarcinoma (Laurén intestinal, diffuse or mixed), grade. Reflex biomarkers on the same block

CT of chest, abdomen and pelvis with intravenous and oral contrast
The first staging test, before endoscopic ultrasound

Distant metastases

M1
Biopsy a metastatic site if doubt remains → **advanced-disease pathways** (chapters 10–12)

No distant metastases

Complete locoregional staging
Choose the tests below by tumor site and by whether the result would change management

If it changes the plan

Endoscopic ultrasound ± FNA
T and N category of non-metastatic tumors. For superficial lesions, **endoscopic resection is the best T-staging**

Esophagus and junction

FDG-PET/CT
Before esophagectomy or definitive chemoradiotherapy: finds occult distant disease in up to 15%. **Not routine in gastric cancer** (diffuse and mucinous tumors are often non-avid)

Stomach, Siewert III

Staging laparoscopy + washings
Before perioperative therapy or gastrectomy (≥cT2 or cN+; NCCN from cT1b). Occult peritoneal disease in about 1 in 5

Selected

Bronchoscopy · ENT
Tumor at or above the carina: airway invasion. Squamous cancer: head-and-neck examination

The host: fitness and nutrition
Performance status and comorbidity; cardiopulmonary reserve before esophagectomy. **Weight loss, body-mass index and sarcopenia** → dietitian and an enteral route if intake fails. Geriatric assessment from 65. Stop smoking and alcohol

Multidisciplinary tumor board: clinical stage → pathway
Superficial (chapter 4) · localized squamous (5) · localized esophageal and junctional adenocarcinoma (6) · resectable gastric (7) · advanced disease (10–12)

Biomarkers: order them with the first biopsy

Adenocarcinoma (esophagus, junction, stomach)
- **HER2** by IHC, ISH if 2+
- **PD-L1** by CPS or TAP
- **MMR/MSI** — also needed before perioperative therapy
- **CLDN18.2** by IHC

Squamous cell carcinoma
- **PD-L1** (CPS, TPS or TAP)
- **MMR/MSI**

When available
- Broad NGS: *NTRK*, *RET*, *BRAF* V600E, tumor mutational burden
- ***DPYD*** genotype before a fluoropyrimidine

NON-NEGOTIABLE — RED FLAGS

- **Histology before any treatment**; CT before endoscopic ultrasound.
- **PET/CT before esophagectomy** or definitive chemoradiotherapy.
- **Laparoscopy with washings** before perioperative therapy in gastric cancer: positive cytology is M1.
- **MMR/MSI before perioperative chemotherapy**; all four biomarkers before first-line therapy of advanced adenocarcinoma.
- **Nutritional screening at the first visit**, and tumor-board review before the first treatment.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Endoscopic ultrasound when CT already shows T3–4 or node-positive disease, or the tumor cannot be traversed.
- PET/CT in gastric cancer; laparoscopy in cT1b disease.
- Breadth of NGS and the use of circulating tumor DNA.
- DPD deficiency testing by *DPYD* genotype or plasma uracil, as locally available.

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2

INFOGRAPHIC

Diagnosis, work-up & biomarkers

ESOPHAGUS STOMACH

HER2

How
IHC first; in-situ hybridisation for IHC 2+

Positive when
IHC 3+, or **IHC 2+ with ISH+** (HER2:CEP17 ≥2). Biopsy: a cluster of 5 or more cells; resection: 10% or more of cells

How often
About 15–20%; up to 30% at the junction and in intestinal type

Opens
Zanidatamab or trastuzumab first line; T-DXd later

PD-L1

How
IHC, scored as CPS, TAP or TPS (see below)

Positive when
≥1 for US labels and ASCO; trial-specific cut-offs (CPS ≥5, TAP ≥5%, CPS ≥10) in Europe

How often
CPS ≥1 in about 80%, ≥5 in 60%, ≥10 in 35% of first-line trial populations

Opens
Nivolumab, pembrolizumab, tislelizumab

MMR / MSI

How
IHC for MLH1, PMS2, MSH2, MSH6, or PCR/NGS

Positive when
Loss of one or more proteins (**dMMR**) or **MSI-high**

How often
3–5% of metastatic and about 8% of resectable gastric cancer; under 1% of esophageal cancers

Opens
Immunotherapy in any line; triggers a Lynch work-up

CLDN18.2

How
IHC (VENTANA CLDN18 43-14A assay)

Positive when
≥75% of tumor cells with moderate-to-strong membranous staining

How often
About 38% of HER2-negative advanced adenocarcinoma

Opens
Zolbetuximab first line; antibody–drug conjugates in trials

NGS and others

How
DNA/RNA panel on tissue; plasma if tissue is short

Positive when
NTRK or *RET* fusion, *BRAF* V600E, TMB ≥10 mut/Mb; EBV by EBER-ISH (about 5–9%)

How often
Each under 1–5%

Opens
Tumor-agnostic drugs in later lines; FGFR2b remains investigational

Three ways to score PD-L1

PD-L1 score	Numerator	Denominator	Used with
CPS combined positive score	PD-L1-stained tumor cells + lymphocytes + macrophages	Viable tumor cells, ×100 (maximum 100)	Pembrolizumab, nivolumab
TAP tumor area positivity	Area covered by PD-L1-stained tumor and immune cells	Total tumor area, as a percentage	Tislelizumab; interchangeable with CPS in NCCN
TPS tumor proportion score	PD-L1-stained tumor cells	Viable tumor cells, as a percentage	Nivolumab in squamous cancer (Europe)

Threshold map: same drug, different label

Drug and setting	FDA · NCCN v4.2026 · ASCO 2026	EMA · ESMO 2026
ADENOCARCINOMA, FIRST LINE		
Nivolumab + chemotherapy	PD-L1 ≥1	CPS ≥5
Pembrolizumab + chemotherapy (HER2−); + trastuzumab (HER2+)	CPS ≥1	CPS ≥1
Tislelizumab + chemotherapy	PD-L1 ≥1	TAP ≥5%
Zanidatamab + tislelizumab + chemotherapy (HER2+)	Any PD-L1 **FDA Aug 2026**	Recommended where available; not yet EMA-approved
Zolbetuximab + chemotherapy	CLDN18.2 ≥75%	CLDN18.2 ≥75%
SQUAMOUS CELL CARCINOMA, FIRST LINE		
Nivolumab + chemotherapy, or + ipilimumab	PD-L1 ≥1	Tumor-cell PD-L1 ≥1%
Pembrolizumab + chemotherapy	CPS ≥1	CPS ≥10
Tislelizumab + chemotherapy	PD-L1 ≥1	TAP ≥5%
RESECTABLE ADENOCARCINOMA		
Durvalumab + FLOT	Label irrespective of PD-L1; NCCN category 1 if CPS/TAP ≥1, 2A if <1	Preferred where available, any PD-L1

ASCO 2026 Good-practice statement

- Test **PD-L1, MMR/MSI, HER2 and CLDN18.2** in every advanced adenocarcinoma, and PD-L1 with MMR/MSI in squamous cancer.
- Results must be available **as early as possible**; consider broad NGS.

ESMO 2026 Four mandatory tests

- HER2 (I, A), PD-L1 (I, A), CLDN18.2 (I, A) and MSI/MMR (II, A) at diagnosis of adenocarcinoma.
- **DPD testing** before 5-FU, capecitabine or S-1 (also an FDA boxed warning since 2025–26). Report HER2 percentage and ISH ratio numerically.

Tissue that can be trusted

- 6–10 biopsies: HER2 and CLDN18.2 are heterogeneous.
- Neutral buffered formalin, 6–72 hours.
- **Re-biopsy at progression** before T-DXd: about 30% lose HER2.
- Keep one block for NGS.

KEY MESSAGE | **No biomarker, no first line.** Four results decide the first-line regimen of adenocarcinoma, two decide it in squamous cancer, and MMR status already matters before the operation.

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DECISION ALGORITHM

3 Anatomy & staging

ESOPHAGUS · STOMACH

AJCC 8th edition remains the staging system for both organs in 2026. The same tumor is staged three times in its life: before treatment, after neoadjuvant therapy, and on the specimen.

cTNM — Clinical stage from endoscopy, CT, PET/CT, endoscopic ultrasound, laparoscopy

ycTNM — Restaging after neoadjuvant therapy, before surgery

ypTNM — Pathology after neoadjuvant therapy: its own stage groups, plus regression grade

pTNM — Pathology when surgery comes first

T category

T	Esophagus and junction	Stomach
Tis	High-grade dysplasia	Carcinoma in situ, intraepithelial
T1a	Lamina propria or muscularis mucosae	Lamina propria or muscularis mucosae
T1b	Submucosa	Submucosa
T2	Muscularis propria	Muscularis propria
T3	Adventitia	Subserosa, without serosal breach
T4a	Pleura, pericardium, azygos vein, diaphragm or peritoneum (resectable)	Serosa (visceral peritoneum)
T4b	Aorta, vertebral body or airway (unresectable)	Adjacent structures or organs

N and M category

N and M	Esophagus and junction	Stomach
N1	1–2 regional nodes	1–2 regional nodes
N2	3–6	3–6
N3	7 or more	N3a 7–15 · N3b 16 or more
Regional nodes	Peri-esophageal cervical nodes down to the celiac nodes	Perigastric nodes and those along the left gastric, common hepatic, splenic and celiac arteries
M1	Distant metastasis, non-regional nodes	Distant metastasis, **positive peritoneal cytology**

Squamous cell carcinoma · clinical stage

Stage	cT	cN	cM
I	T1	N0–1	M0
II	T2 / T3	N0–1 / N0	M0
III	T3 / T1–3	N1 / N2	M0
IVA	T4 / any T	N0–2 / N3	M0
IVB	Any T	Any N	M1

Esophageal and junctional adenocarcinoma

Stage	cT	cN	cM
I	T1	N0	M0
IIA	T1	N1	M0
IIB	T2	N0	M0
III	T2 / T3–4a	N1 / N0–1	M0
IVA	T1–4a / T4b / any T	N2 / N0–2 / N3	M0
IVB	Any T	Any N	M1

Gastric cancer · clinical stage

Stage	cT	cN	cM
I	T1–2	N0	M0
IIA	T1–2	N1–3	M0
IIB	T3–4a	N0	M0
III	T3–4a	N1–3	M0
IVA	T4b	Any N	M0
IVB	Any T	Any N	M1

The 2-cm rule at the junction

A tumor that involves the junction with its epicenter **within 2 cm** of it in the proximal stomach is staged as **esophageal**. Epicenter more than 2 cm below the junction, or a cardia tumor not involving the junction: staged as **gastric**

Tis – T1a — Endoscopic resection chapter 4

T1b – T2 N0 — Surgery; multimodality if high-risk chapters 5–7

T2 or more, or N+ — Perioperative chemotherapy or chemoradiotherapy + surgery chapters 5–7

T4b, unresectable — Definitive chemoradiotherapy or systemic therapy chapters 5, 10, 11

M1 — Systemic therapy, palliation chapters 10–12

NON-NEGOTIABLE — RED FLAGS

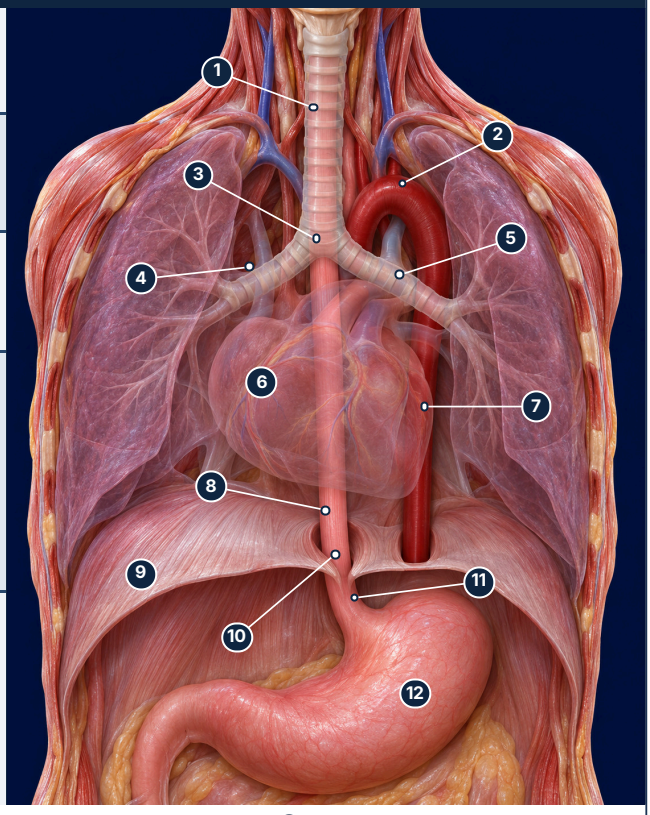
- **Stage with AJCC 8th edition** and state the histology: squamous and adenocarcinoma have different stage groups.
- **Apply the 2-cm rule** to every junctional tumor and record the Siewert type.
- **Positive peritoneal cytology is M1** in gastric cancer.
- **Adequate nodal yield:** at least 15 nodes for esophagectomy and 16 for gastrectomy (30 or more is better).
- **After neoadjuvant therapy** report ypTNM, margin (R) status and tumor regression grade.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- cT2N0 esophageal cancer: up to half are under-staged; surgery first or neoadjuvant therapy.
- Nodal staging: endoscopic ultrasound, PET/CT and CT disagree often; biopsy the node that would change the plan.
- T4a versus T4b: resectability is a surgical judgment at the tumor board.
- PET response after induction chemotherapy as a guide to further therapy.

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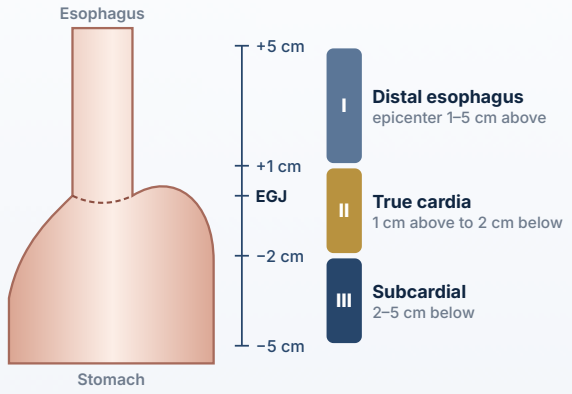
3D The esophagus in the mediastinum, anterior view



1. **Trachea.** Its soft back wall rests on the esophagus.
2. **Aortic arch**, crossing on the left.
3. **Carina**, about 25 cm from the incisors.
4. **Right main bronchus.**
5. **Left main bronchus**, crossing in front of the esophagus.
6. **Heart.** The left atrium lies directly in front.
7. **Descending aorta**, alongside the lower esophagus.
8. **Esophagus**, seen through the heart.
9. **Diaphragm.**
10. **Esophageal hiatus.**
11. **Junction**, about 40 cm; the Z-line.
12. **Stomach.**

Distances are from the incisors at endoscopy; a tumor belongs to the segment that holds its **epicenter**. Airway, aorta and spine lie within millimeters of a wall that has no serosa: invasion of any of them is T4b. The scale is approximate.

Siewert classification of junctional adenocarcinoma



Defined by the tumor epicenter relative to the anatomical junction.

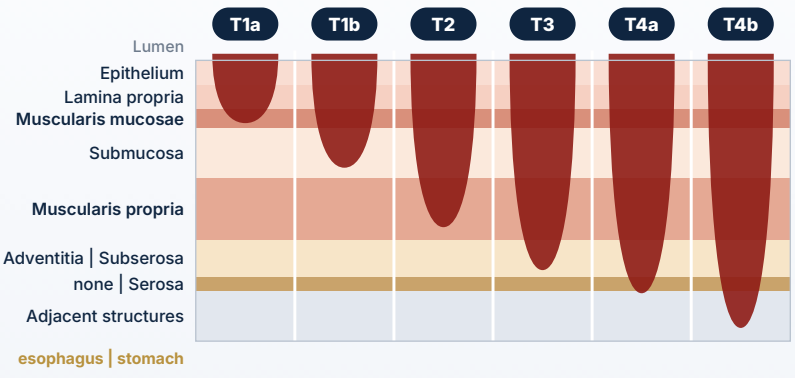
From Siewert type to operation

Siewert	Behaves like	Usual operation
Type I	Distal esophageal adenocarcinoma	Esophagectomy (Ivor Lewis) with two-field lymphadenectomy
Type II	True cardia	Esophagectomy **or** extended total gastrectomy Negotiable
Type III	Subcardial gastric cancer	Total gastrectomy with distal esophageal resection and D2

After neoadjuvant therapy, the pathology report must state

- **ypT and ypN**, with nodes positive over nodes examined.
- **Tumor regression grade** (Becker, Mandard or modified Ryan) and the percentage of residual tumor.
- **R status** and the circumferential margin: positive at 0 mm (CAP) or within 1 mm (Royal College).
- **Lymphovascular and perineural invasion**; HER2, MMR and PD-L1 if not done on the biopsy.

Depth of invasion defines the T category



Left of the bar: esophagus; right: stomach. T4a: pleura, pericardium, azygos vein, diaphragm or peritoneum in the esophagus; serosa in the stomach. T4b: aorta, vertebra or airway; adjacent organs.

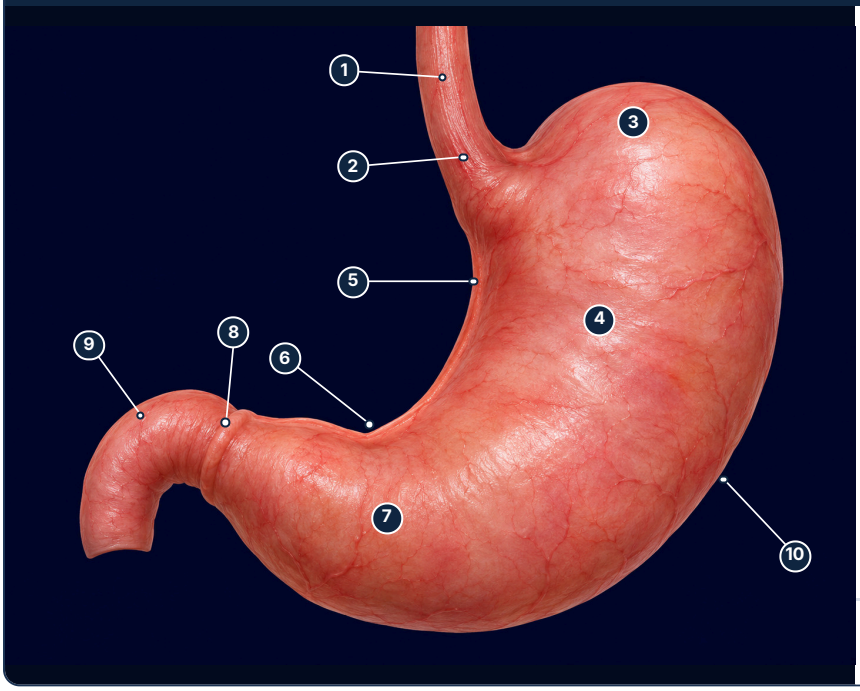
Why the anatomy dictates the treatment

- **No serosa on the esophagus:** early spread to the aorta, airway and pericardium; radial margins are narrow.
- **Submucosal lymphatics run lengthwise:** skip metastases and long margins; a T1b tumor already carries nodal risk.
- **Upper and middle third** drain to the neck and mediastinum; **lower third and junction** to the left gastric and celiac nodes.
- **Stomach:** perigastric nodes first, then along the celiac branches: the logic of D2 (chapter 8).
- **Serosal breach (T4a)** opens the peritoneal cavity: laparoscopy and cytology.

KEY MESSAGE | Three measurements settle most decisions: **distance from the incisors**, **relation of the epicenter to the junction**, and **depth of invasion**. Record all three at the first endoscopy.

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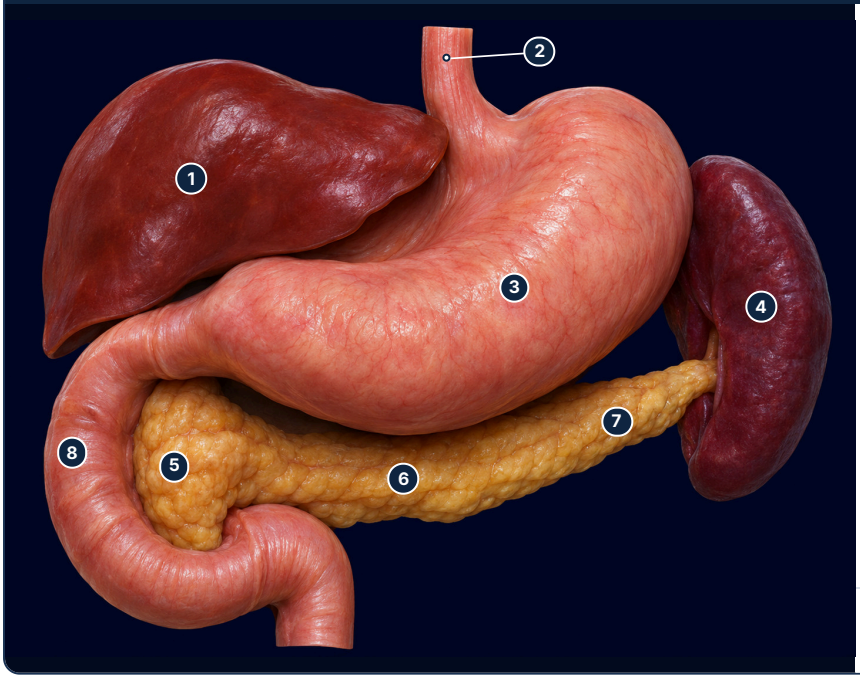
3D Stomach: surface landmarks, anterior view



1. **Abdominal esophagus.** The short last segment, below the hiatus.
2. **Junction and cardia.** The zone of Siewert tumors; the 2-cm rule picks the staging system.
3. **Fundus.** Above the level of the junction; short gastric vessels (station 4sa).
4. **Body.** Oxyntic mucosa: acid and intrinsic factor.
5. **Lesser curvature.** Left and right gastric arteries; stations 1, 3 and 5.
6. **Angular incisure.** Body meets antrum; atrophy and metaplasia are often most advanced here: one of the five Sydney biopsies.
7. **Antrum.** Pyloric mucosa with gastrin cells; with the lesser curvature, the classic site of intestinal-type cancer.
8. **Pylorus.** Suprapyloric (5) and infrapyloric (6) nodes.
9. **Duodenal bulb.** Divided just beyond the pylorus in a distal or total gastrectomy.
10. **Greater curvature.** Gastroepiploic arcades; stations 4sb and 4d.

JGCA divides each curvature into three equal parts to define the **upper (U), middle (M) and lower (L)** thirds: the vocabulary of every endoscopy and pathology report.

3D Stomach: the neighbors that decide resectability



1. **Liver**, drawn lifted: its left lobe normally overlies the lesser curvature.
2. **Esophagus** at the hiatus of the diaphragm.
3. **Stomach.** Its back wall faces the lesser sac.
4. **Spleen**, tied to the fundus by the gastrosplenic ligament.
5. **Head of the pancreas**, in the curve of the duodenum.
6. **Body of the pancreas**, behind the stomach; the splenic artery runs along its upper border.
7. **Tail of the pancreas**, reaching the splenic hilum.
8. **Duodenum.**

WHICH WALL, WHICH NEIGHBOR

Front wall: left lobe of the liver, diaphragm, abdominal wall.
Back wall: lesser sac, then the pancreas, splenic vessels, left adrenal and kidney.
Greater curvature: transverse colon and mesocolon, spleen.

Organs are drawn apart for clarity. The transverse colon and its mesocolon, which lie below the greater curvature, are not shown.

What counts as T4b in the stomach (AJCC 8th)

- **Adjacent structures:** spleen, transverse colon, liver, diaphragm, pancreas, abdominal wall, adrenal gland, kidney, small intestine and retroperitoneum.
- **Not T4b:** spread within the wall into the esophagus or duodenum; T is set by the deepest invasion at any of these sites.
- **Serosa breached without an adjacent organ:** T4a, and the peritoneal cavity is open: staging laparoscopy with cytology.

The ligaments carry the vessels and the nodes

- **Lesser omentum:** left and right gastric arteries; stations 1, 3 and 5; its free edge holds 12a.
- **Greater omentum** (gastrocolic ligament): the gastroepiploic arcades; stations 4sb, 4d and 6.
- **Gastrosplenic ligament:** short gastric vessels (4sa), leading to the splenic hilum (10).
- **Behind the stomach:** the celiac trunk; stations 7, 8a, 9 and 11 follow its branches (chapter 8).

KEY MESSAGE | Name the tumor by **third and curvature**, then ask which neighbor it touches: the first fixes the operation and the nodal field, the second decides whether the resection can be R0.

3D plates: computer-generated educational models supplied by the author. They show normal anatomy, are not patient images and are not to scale.

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DECISION ALGORITHM

4 Early disease: endoscopic therapy

ESOPHAGUS · STOMACH

Superficial lesion (Tis or T1) at endoscopy
Barrett dysplasia, early squamous cancer or early gastric cancer

Characterize before touching
High-definition white light with virtual or dye chromoendoscopy (Lugol iodine for squamous mucosa). Paris morphology, size, ulceration. Endoscopic ultrasound or CT only if deep invasion or nodes are suspected

Is the lesion endoscopically resectable?
No sign of deep submucosal invasion; no suspicious nodes; an expert endoscopist available

No

Surgery, or chemoradiotherapy for squamous cancer
Chapters 5–7

Yes

En-bloc endoscopic resection: the resection is the staging
ESD for squamous and gastric lesions and for Barrett lesions over 15–20 mm or with suspected submucosal invasion; **EMR** for smaller Barrett lesions

Expert pathology of the pinned, oriented specimen
Depth in micrometers · lateral and deep margins · differentiation · lymphovascular invasion · tumor budding · ulceration

Squamous cell carcinoma

Curative
m1–m2 (epithelium, lamina propria), no lymphovascular invasion, clear margins

Individualize
m3 or sm1 up to 200 µm, well or moderately differentiated, no lymphovascular invasion: observe, or add chemoradiotherapy or esophagectomy

Not curative
Deeper submucosa, lymphovascular invasion, poor differentiation or positive deep margin → **esophagectomy or chemoradiotherapy**

Barrett adenocarcinoma

Curative
T1a; or T1b **sm1 up to 500 µm** with grade 1–2, no lymphovascular invasion, clear margins

Then
Ablate all remaining Barrett epithelium (radiofrequency) and maintain high-dose acid suppression

Not curative
Deeper submucosa or any high-risk feature → **esophagectomy**

Gastric cancer (JGCA, eCura)

Curative: eCura A or B
- Differentiated T1a, no ulcer, any size
- Differentiated T1a, ulcerated, up to 3 cm
- Undifferentiated T1a, no ulcer, up to 2 cm
- Differentiated T1b-sm1 (under 500 µm), up to 3 cm

Then
Test and treat *H. pylori*; endoscopy every 6–12 months

Not curative: eCura C-2
Gastrectomy with lymphadenectomy (D1+ or D2)

EMR or ESD?

	EMR (cap or band)	ESD
En-bloc resection	Lesions up to 15–20 mm	Any size
Margins and recurrence	Piecemeal above that size: margins unreadable, more recurrence	R0 in over 90% in expert centers; recurrence very low
Price	Quick, low perforation risk	Longer; perforation 2–5%; stricture after wide esophageal resection
Best use	Small nodular Barrett lesions	Squamous and gastric cancer; larger Barrett lesions or suspected submucosal invasion

Endoscopic signs of deep submucosal invasion

- **Excavated or ulcerated** morphology (Paris 0-III), or a depressed area within a nodule.
- **Non-lifting** after submucosal injection, without previous biopsy scarring.
- **Fold convergence, fusion or rigidity** of the wall.
- **Magnifying narrow-band imaging:** lost surface pattern, large irregular vessels (type B3).

NON-NEGOTIABLE — RED FLAGS

- **Resect, do not ablate, a visible lesion**: ablation destroys the T-staging.
- **En-bloc specimen, pinned and oriented**, read by an expert gastrointestinal pathologist.
- **Lymphovascular invasion, deep submucosal invasion or a positive deep margin:** back to the tumor board.
- **After Barrett resection, eradicate the residual Barrett segment.**
- **After gastric ESD, eradicate *H. pylori*** and keep endoscopic surveillance: metachronous cancers are common.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- m3 or sm1 squamous cancer: observation, chemoradiotherapy or surgery.
- Low-risk sm1 Barrett adenocarcinoma: endoscopic follow-up versus esophagectomy.
- Non-curative gastric ESD in a frail patient with a low eCura score: observe.
- Circumferential esophageal ESD: stricture risk; steroid prophylaxis.

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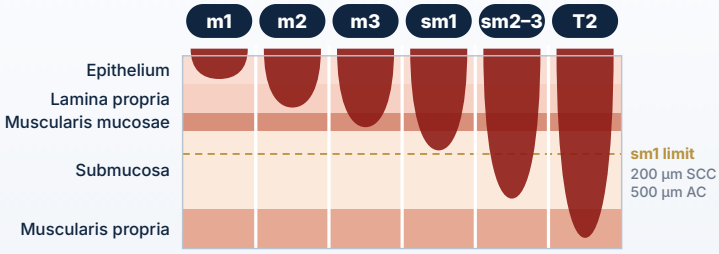
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INFOGRAPHIC

Early disease: endoscopic therapy

ESOPHAGUS · STOMACH

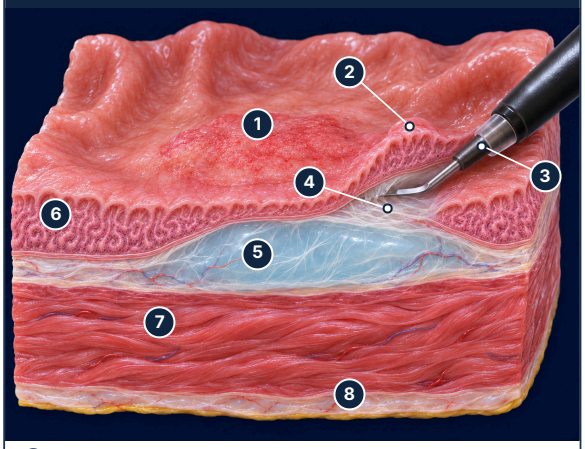
How deep is too deep for the endoscope



Nodal metastasis by depth (approximate, pooled surgical series) and what it means for endoscopic resection

Depth	m1	m2	m3	sm1	sm2–3	T2
Squamous: nodal metastasis	0–3%	0–3%	about 10%	10–25%	25–50%	>50%
Barrett adenocarcinoma	1–2%	1–2%	1–2%	about 2% if low-risk	15–25%	higher
Gastric cancer	2–5% overall; near 0% within the criteria			about 10%	15–20%	higher
Endoscopic resection	Curative	Curative	Individualize (squamous)	Individualize	Not adequate	Not adequate

3D Endoscopic submucosal dissection (ESD)



1. **Early cancer** confined to the mucosa.
2. **Mucosal flap**, lifted with the lesion.
3. **ESD knife.**
4. **Dissection plane** in the deep submucosa.
5. **Fluid cushion** injected under the lesion.
6. **Mucosa** and its glands.
7. **Muscularis propria**, left intact.
8. **Serosa.**

The lesion comes out in one piece, so the pathologist can measure depth, margins and vessel invasion. Shown in the stomach; the technique is the same in the esophagus.

Gastric ESD: the eCura curability system (JGCA)

eCura	Meaning	Next step
A	Curative: within the absolute indication, en bloc, clear margins, no lymphovascular invasion	Endoscopy every 6–12 months
B	Curative for the expanded indication	Endoscopy plus ultrasound or CT every 6–12 months
C-1	Differentiated, would be A or B, but piecemeal or positive lateral margin	Repeat ESD, coagulate, or observe closely
C-2	Non-curative: all other cases	Gastrectomy with lymphadenectomy

How much risk is left behind?

eCura risk score after a non-curative gastric ESD	Points	Nodal metastasis
Lymphatic invasion	3	
Tumor over 30 mm · positive deep margin · venous invasion · sm2 or deeper	1 each	
Low risk	0–1	2.5%
Intermediate risk	2–4	6.7%
High risk	5–7	22.7%

>90%
en-bloc resection with ESD in expert centers
curative resection in about 80–90%

97%
5-year overall survival after ESD for expanded-indication gastric cancer
JCOG0607

2–5%
perforation; delayed bleeding about 5%
stricture after wide esophageal ESD

92.6%
3-year survival, esophageal squamous cT1b after endoscopic resection + selective chemoradiotherapy
JCOG0508

After a curative resection

- **Esophagus:** endoscopy every 3 months in year 1, every 6 months in year 2, then yearly.
- **Barrett:** radiofrequency ablation sessions until complete eradication of intestinal metaplasia, then surveillance.
- **Stomach:** endoscopy every 6–12 months; eradicate *H. pylori*; metachronous cancers arise at about 3% a year.
- **Squamous:** stop smoking and alcohol; look for a second aerodigestive primary.

ESMO 2026 Early disease

- Carefully selected stage I adenocarcinoma: endoscopic or surgical resection alone may be curative (III, B).

NCCN v4.2026 · JGCA · JCOG0502

- **NCCN:** depth of submucosal invasion in T1b is now **quantified** in the pathology report.
- **JGCA:** ESD is the standard technique for early gastric cancer within the criteria.
- **JCOG0502** (non-randomized, n = 368), cT1b N0 squamous cancer: 5-year survival 86.5% after esophagectomy, 85.5% after chemoradiotherapy.

KEY MESSAGE | Endoscopic resection is both the **best staging test** and, when the specimen meets the criteria, the **cure**. The decision is made on the pathology report, not at the endoscopy.

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DECISION ALGORITHM

5 Esophageal squamous cell carcinoma: localized disease

ESOPHAGUS · SQUAMOUS

Squamous cell carcinoma of the esophagus, M0

PET/CT and endoscopic ultrasound done · bronchoscopy if at or above the carina · nutrition secured · tumor-board decision

cT1b–T2 N0, low risk

Esophagectomy
Under 3 cm and well differentiated. Definitive chemoradiotherapy if unfit or surgery is declined

Pathology
Node-positive or margin-positive: adjuvant treatment at the tumor board

cT2 high risk, cT3–T4a or N+ · resectable and fit

Neoadjuvant chemoradiotherapy (CROSS)
Carboplatin AUC 2 + paclitaxel 50 mg/m² weekly ×5 with **41.4 Gy in 23 fractions** Cat 1 · I, A

Restage at 4–6 weeks
PET/CT; endoscopy with biopsies if organ preservation is discussed

Esophagectomy
Two-field lymphadenectomy, 15 nodes or more

Residual disease (ypT1 or more, or ypN+), R0
Adjuvant nivolumab for 1 year Cat 1 · I, A. ypT0 N0: surveillance

Clinical complete response and the patient wishes to keep the esophagus
Active surveillance in a strict protocol (SANO) Negotiable; ESMO still recommends surgery

Cervical tumor · unfit · declines surgery

Definitive chemoradiotherapy
50–50.4 Gy in 25–28 fractions with FOLFOX, carboplatin–paclitaxel or cisplatin–5-FU III, B

Response at 8–12 weeks
PET/CT + endoscopy with biopsies

Complete response
Endoscopy and CT every 3 months for 2 years

Persistent or recurrent local disease
Salvage esophagectomy in an expert center IV, C

cT4b, unresectable

Definitive chemoradiotherapy
Adjacent structures involved (aorta, vertebral body, trachea), no fistula

Tracheobronchial invasion with a threatened fistula
Consider chemotherapy alone first; avoid a stent before chemoradiotherapy when possible

Good response
Re-discuss resection at the tumor board

No pembrolizumab with chemoradiotherapy
KEYNOTE-975
ESMO 2026 · Negative

Where radiotherapy cannot be given, or by regional practice: neoadjuvant docetaxel, cisplatin and 5-FU (DCF) ×3, then esophagectomy

JCOG1109 NExT: 3-year survival 72.1% with DCF versus 62.6% with cisplatin–5-FU; febrile neutropenia 16% — primary G-CSF Negotiable

Who is fit for chemoradiotherapy and surgery? The CROSS yardstick

- **Stage:** T1 N1 or T2–3 N0–1, M0; tumor up to 8 cm long and 5 cm wide.
- **Patient:** age 18–75, WHO performance status 0–2, weight loss of 10% or less.
- **Organs:** adequate marrow, renal, hepatic, cardiac and pulmonary function for a thoracotomy.
- Outside these limits the evidence thins: decide at the tumor board.

Clinical complete response: the SANO definition

- **4–6 weeks** after chemoradiotherapy: endoscopy with bite-on-bite biopsies.
- **12 weeks:** PET/CT, endoscopy with bite-on-bite biopsies, endoscopic ultrasound with needle sampling of suspicious nodes.
- **All negative:** surveillance every 3 months in year 1, every 4 months in year 2, every 6 months in year 3, then yearly to year 5.
- **Regrowth without metastases:** esophagectomy.

NON-NEGOTIABLE — RED FLAGS

- **Bronchoscopy** for tumors at or above the carina before any curative plan.
- **Secure nutrition first**; in surgical candidates prefer a feeding jejunostomy or nasoenteral tube to a gastrostomy, which can compromise the gastric conduit.
- **No dose escalation above 50.4 Gy** in the thoracic esophagus (ARTDECO, CONCORDE).
- **Adjuvant nivolumab only** for residual disease after chemoradiotherapy and R0 resection.
- **Do not add pembrolizumab** to definitive chemoradiotherapy.
- **Surveillance after a clinical complete response** needs endoscopy with bite-on-bite biopsies, endoscopic ultrasound and PET/CT on schedule, or it is not surveillance.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Chemoradiotherapy or DCF chemotherapy before surgery.
- Surgery versus surveillance after a clinical complete response (SANO: 2-year survival 74% vs 71%).
- Surgery versus definitive chemoradiotherapy in responders: better local control, no proven survival gain.
- Adjuvant nivolumab when PD-L1 is below 1.
- A higher dose (60–66 Gy) for cervical tumors.
- Consolidation immunotherapy after definitive chemoradiotherapy (SKYSCRAPER-07): not licensed.

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INFOGRAPHIC

5 Esophageal squamous cell carcinoma: localized disease

ESOPHAGUS · SQUAMOUS

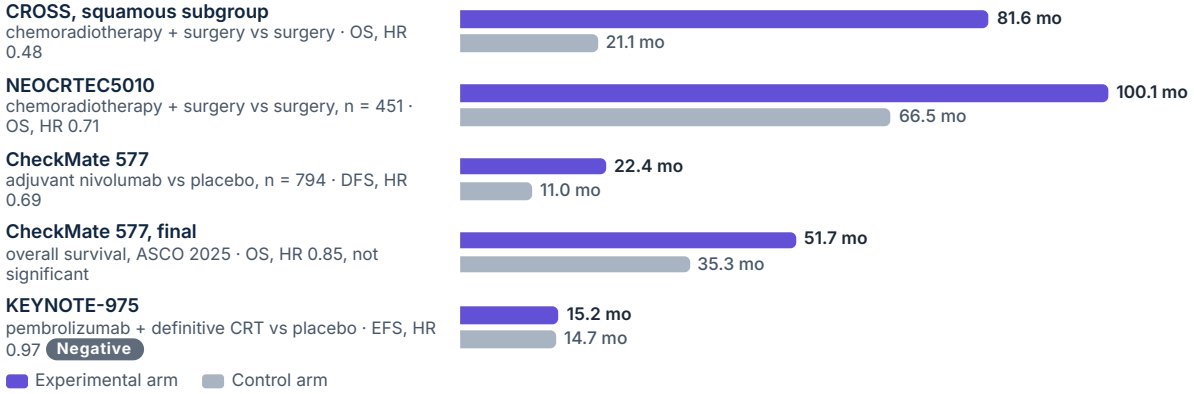
49% pathological complete response after CROSS in squamous cancer
23% in adenocarcinoma

46% 10-year survival with CROSS + surgery in squamous cancer
versus 23% with surgery alone

74% 2-year survival under active surveillance after a clinical complete response
71% with standard surgery · SANO

72% 3-year survival with neoadjuvant DCF
63% with cisplatin–5-FU · JCOG1109

Median survival in the pivotal trials (months)



Definitive chemoradiotherapy: five questions, five answers

Question	Trial	Answer
Chemoradiotherapy or radiotherapy alone?	RTOG 85-01	5-year survival 26% vs 0%: concurrent chemotherapy is mandatory
Does a higher dose help?	INT 0123 · ARTDECO · CONCORDE	64.8, 61.6 or 66 Gy no better than 50–50.4 Gy (ARTDECO: 3-year local progression-free survival 73% vs 70%, not significant)
Which chemotherapy?	PRODIGE 5	FOLFOX equals cisplatin–5-FU (PFS 9.7 vs 9.4 months); weekly carboplatin–paclitaxel widely used
Add surgery after a response?	FFCD 9102 · Stahl	Better local control, no survival gain in responders
Add immunotherapy?	KEYNOTE-975 · SKYSCRAPER-07	Concurrent pembrolizumab: no. Atezolizumab after CRT: OS HR 0.69, descriptive only

Weeks 1–5 CROSS: weekly carboplatin–paclitaxel with 41.4 Gy

Weeks 9–11 Restaging: PET/CT ± endoscopy

Weeks 10–13 Esophagectomy

Within 4–16 weeks Adjuvant nivolumab if residual disease: 240 mg every 2 weeks ×16 weeks, then 480 mg every 4 weeks to 1 year

ESMO 2026 Guideline and congress

- CROSS chemoradiotherapy, restaging and surgery (I, A); **surgery still recommended after a clinical complete response.**
- Adjuvant nivolumab for 1 year when residual disease is found (I, A).
- **KEYNOTE-975:** pembrolizumab with definitive chemoradiotherapy did not improve event-free or overall survival and raised discontinuations (34% vs 27%).
- **SKYSCRAPER-07** (Ann Oncol 2026): tiragolumab + atezolizumab failed; atezolizumab alone, 24-month survival 69% vs 59%.

ASCO 2025–26 What ASCO added

- **CheckMate 577 final OS:** 51.7 vs 35.3 months, not significant; 5-year survival 46% vs 41%; benefit mainly when PD-L1 CPS is 1 or more.
- **SCIENCE** (ASCO 2026, China): neoadjuvant sintilimab with chemotherapy or chemoradiotherapy; pathological complete response reported, event-free survival immature (Regional).
- The locally advanced esophageal guideline is in revision.

KEY MESSAGE | Squamous cancer is the **radiotherapy-sensitive** histology: chemoradiotherapy first, then surgery for the fit, definitive chemoradiotherapy for the rest, and nivolumab only for those with residual disease after both.

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DECISION ALGORITHM

6 Esophageal & junctional adenocarcinoma: localized disease

ESOPHAGUS / GEJ · ADENOCARCINOMA

Adenocarcinoma of the esophagus or junction (Siewert I–III), M0

MMR/MSI and PD-L1 reported · PET/CT done · laparoscopy for Siewert III and bulky type II · tumor-board decision

cT1b–T2 N0, low risk

Esophagectomy
Under 3 cm, well differentiated. T1a: chapter 4

Pathology
Upstaged to node-positive: adjuvant therapy at the tumor board

cT2 high risk or more, or cN+ · resectable · fit for a triplet

Perioperative FLOT ×4
Docetaxel, oxaliplatin, leucovorin and 5-FU every 2 weeks (Cat 1) (I, A)

Add durvalumab (D-FLOT) for junctional tumors
NCCN: preferred when PD-L1 CPS or TAP is 1 or more (category 1 junction, 2A esophageal body); 2A when below 1 or diffuse type. ESMO: preferred where available ASCO ESMO

Surgery 4–6 weeks after cycle 4
Siewert I–II: esophagectomy. Siewert III: total gastrectomy with D2

FLOT ×4 again
With durvalumab if started, then **durvalumab every 4 weeks ×10**

Not fit for FLOT

Neoadjuvant chemoradiotherapy (CROSS)
41.4 Gy with weekly carboplatin–paclitaxel

Esophagectomy

Residual disease, R0
Adjuvant nivolumab for 1 year; ESMO: when PD-L1-positive (I, B)

Alternative
Perioperative FOLFOX or CAPOX doublet

Unresectable or inoperable

Definitive chemoradiotherapy
50–50.4 Gy with FOLFOX or carboplatin–paclitaxel

Reassess
Resection if it becomes feasible; otherwise surveillance or systemic therapy

Mismatch-repair deficient or MSI-high tumor (about 8% in the stomach, under 1% in the esophagus)
Chemotherapy alone adds little. Options: **neoadjuvant or perioperative immunotherapy** (nivolumab + ipilimumab then nivolumab; pembrolizumab; tremelimumab + durvalumab; dostarlimab), or D-FLOT. Surgery remains the standard after response; watch-and-wait is investigational Negotiable

FLOT or CROSS: the comparison in one view

	Perioperative FLOT ± durvalumab	CROSS chemoradiotherapy
Aim	Systemic control: distant relapse is the main failure	Local control and a clear radial margin
Length	8 weeks before and 8 weeks after surgery; durvalumab to 1 year	5 weeks before surgery
Pathological complete response	About 17% (ESOPEC)	About 10% in ESOPEC; 23% in CROSS
3-year survival, ESOPEC	**57.4%**	50.7%
Immunotherapy	Durvalumab within the regimen (junction, stomach)	Nivolumab afterwards if residual disease
Best for	Fit patients, performance status 0–1	Unfit for a triplet; squamous histology

Fit for FLOT? A bedside checklist

- Performance status 0–1; able to eat or fed enterally.
- Adequate marrow, renal, hepatic and cardiac function.
- No neuropathy of grade 2 or more.
- *DPYD* result available before cycle 1.
- Central line and a 24-hour pump are feasible.
- G-CSF plan, above all after 65.

NON-NEGOTIABLE — RED FLAGS

- **MMR/MSI and PD-L1 before the first cycle.**
- **FLOT is for fit patients**: performance status 0–1, primary G-CSF considered, neuropathy watched.
- **Plan the postoperative half** from the start: fewer than half of patients complete all eight cycles.
- **Do not add radiotherapy** to perioperative chemotherapy as routine (TOPGEAR).
- **Adjuvant nivolumab belongs after chemoradiotherapy**, not after FLOT.
- **No anti-HER2 or zolbetuximab** perioperatively outside trials.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Durvalumab + FLOT when PD-L1 is below 1 or the tumor is diffuse-type (no survival gain in that subgroup).
- D-FLOT for adenocarcinoma of the esophageal body, which MATTERHORN did not enroll.
- Esophagectomy or extended total gastrectomy for Siewert II.
- Surgery first or perioperative therapy for cT2 N0.
- Older patients: dose-modified FLOT, a doublet, or CROSS.
- MSI-high: immunotherapy alone, and organ preservation after a complete response.

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INFOGRAPHIC

6 Esophageal & junctional adenocarcinoma: localized disease

ESOPHAGUS / GEJ · ADENOCARCINOMA

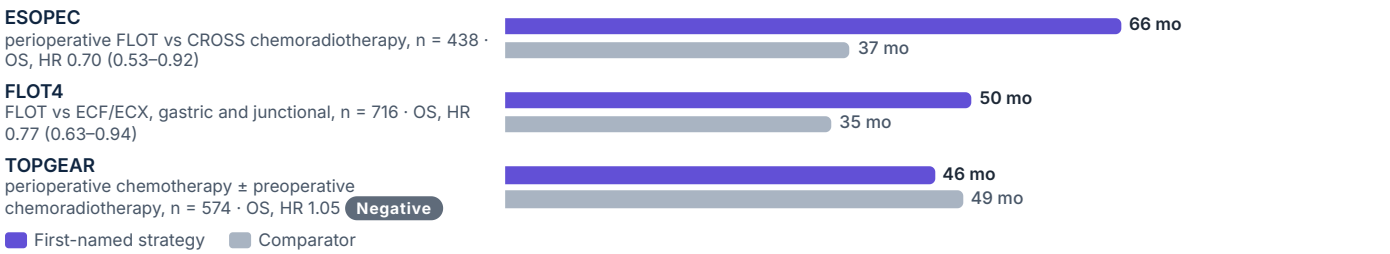
66 mo
median survival with perioperative FLOT in esophageal adenocarcinoma
37 months with CROSS · ESOPEC

0.78
hazard ratio for death, durvalumab + FLOT vs FLOT
p = 0.021 · MATTERHORN final

69 %
3-year survival with durvalumab + FLOT
62% with FLOT alone

19 %
pathological complete response with D-FLOT
7% with FLOT

Median overall survival (months)



THE D-FLOT SCHEDULE (MATTERHORN)

Weeks 1–8
FLOT ×4, every 2 weeks; durvalumab 1,500 mg on day 1 of weeks 1 and 5

Weeks 12–14
Restage, then surgery with adequate lymphadenectomy

From 4–12 weeks after surgery
FLOT ×4 with durvalumab ×2

Maintenance
Durvalumab 1,500 mg every 4 weeks ×10

FLOT in practice

- Docetaxel 50, oxaliplatin 85, leucovorin 200 and 5-FU 2,600 mg/m² over 24 hours, day 1 every 2 weeks.
- Grade 3–4 neutropenia in about half: consider primary G-CSF, especially over 65.
- Check *DPYD*; watch diarrhea and cumulative neuropathy.
- Grade 3–4 events were the same with or without durvalumab (71.6% vs 71.2%).

MSI-high localized adenocarcinoma

- **NEONIPIGA** (nivolumab + ipilimumab): pathological complete response 59%.
- **INFINITY** (tremelimumab + durvalumab): 60%.
- **DEWI** (dostarlimab, watch-and-wait): 62.5% without surgery at 1 year ESMO 2026.
- Perioperative chemotherapy alone: little or no benefit in retrospective analyses.

What did not change the standard, and why it matters

Trial	Comparison	Result	Message
Neo-AEGIS	Perioperative chemotherapy (mostly pre-FLOT) vs CROSS	3-year survival 55% vs 57%, HR 1.03	Equivalent in the pre-FLOT era; superseded by ESOPEC
TOPGEAR	Adding 45 Gy preoperatively to perioperative chemotherapy	pCR 17% vs 8%; no survival or PFS gain	More response, no benefit: no routine radiotherapy
CheckMate 577	Adjuvant nivolumab after CRT + surgery (71% adenocarcinoma)	DFS doubled (22.4 vs 11.0 months); final OS HR 0.85, not significant	Use after CRT when residual disease, above all if PD-L1 ≥1
KEYNOTE-585	Perioperative pembrolizumab + chemotherapy	pCR 13% vs 2%; EFS HR 0.81, not significant	pCR alone did not translate; durvalumab + FLOT did

ESMO 2026

- **FLOT is preferred over CROSS** for esophageal and junctional adenocarcinoma (I, A) (MCBS A).
- **Durvalumab + FLOT** preferred where available (I, A); EC approval March 2026.
- Chemoradiotherapy only for those unsuitable for FLOT.

ASCO 2025–26

- MATTERHORN event-free survival (2025) led to approval FDA 25 Nov 2025.
- NCCN v4.2026 (Sep 2026) raised D-FLOT to category 2A for PD-L1 below 1 and for diffuse-type junctional tumors.
- ASCO guideline for locally advanced esophageal cancer (2020): update in development.

KEY MESSAGE | For adenocarcinoma the systemic disease decides the outcome: **perioperative FLOT, with durvalumab at the junction,** has replaced chemoradiotherapy as the first choice for fit patients.

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DECISION ALGORITHM

7 Resectable gastric cancer

STOMACH

Gastric adenocarcinoma, M0

CT ± endoscopic ultrasound · **staging laparoscopy with cytology negative** · MMR/MSI and PD-L1 reported · nutrition optimized · tumor-board decision

cT1 N0

T1a within criteria
Endoscopic submucosal dissection (chapter 4)

T1b, or T1a beyond criteria
Gastrectomy with D1+ or D2 lymphadenectomy

Pathology
pN+ or pT3–4: adjuvant chemotherapy

cT2 or more, or cN+ · perioperative strategy

Durvalumab + FLOT ×4 (D-FLOT)
Preferred; NCCN category 1 when PD-L1 CPS or TAP is 1 or more, 2A when below 1 (I, A) FDA Nov 2025

FLOT ×4
Where durvalumab is unavailable or contraindicated (Cat 1)

Unfit for a triplet
Perioperative FOLFOX or CAPOX

D2 gastrectomy
4–6 weeks after the last preoperative cycle; 16 nodes or more, R0

Same regimen ×4 after surgery
D-FLOT: then durvalumab every 4 weeks ×10

Surgery first

D2 gastrectomy
East Asian standard; also bleeding or obstruction, or selected cT2 N0

Stage II–III after D2
- **CAPOX for 6 months** (CLASSIC); 4 months non-inferior (LOMAC) ESMO 2026
- East Asia: S-1 for 1 year (stage II); S-1 + docetaxel (stage III)

Less than D2, or R1
Fluoropyrimidine **chemoradiotherapy**, 45 Gy (INT-0116)

pT1–2 N0
Surveillance

Mismatch-repair deficient or MSI-high
Avoid chemotherapy alone. Neoadjuvant or perioperative immunotherapy (NCCN options: nivolumab + ipilimumab then nivolumab; pembrolizumab; tremelimumab + durvalumab; dostarlimab), D-FLOT, or surgery first; decide at the tumor board Negotiable

Positive cytology or peritoneal deposits at laparoscopy
Stage IV: systemic therapy (chapters 10 and 12). Conversion surgery only after a durable response and a tumor-board decision

Weeks 1–8
D-FLOT ×4: FLOT every 2 weeks, durvalumab 1,500 mg in weeks 1 and 5

Weeks 12–14
Restaging CT, then D2 gastrectomy

4–12 weeks after surgery
D-FLOT ×4 again

To 1 year
Durvalumab 1,500 mg every 4 weeks ×10

Staging laparoscopy: why it is not optional

- **Who:** every potentially resectable tumor of stage cT1b or more (NCCN); stage IB–III (ESMO III, B).
- **What it finds:** peritoneal disease that CT missed in roughly one patient in five.
- **Washings:** positive cytology is M1 even when the peritoneum looks clean.
- **Repeat** after systemic therapy if conversion surgery is discussed.

Adjuvant treatment by pathology

Pathology after upfront D2	Adjuvant treatment
pT1, or pT2 N0	Surveillance
Stage II	CAPOX for 4–6 months, or S-1 for 1 year
Stage III	CAPOX or SOX for 4–6 months; East Asia: S-1 + docetaxel
Less than D2, or R1	Fluoropyrimidine chemoradiotherapy, 45 Gy
MSI-high, stage II	Surveillance is reasonable: chemotherapy adds little

NON-NEGOTIABLE — RED FLAGS

- **Staging laparoscopy with washings** before perioperative therapy or gastrectomy.
- **D2 lymphadenectomy, 16 nodes or more, in a high-volume center**; spleen and pancreas preserved.
- **MMR/MSI before chemotherapy.**
- **R0 resection**: frozen section of the margins in diffuse-type tumors.
- **No adjuvant radiotherapy after an R0 D2 resection** (ARTIST 2, CRITICS).
- **No HIPEC outside trials** (PERISCOPE II, GASTRIPEC-I).
- **Nutritional prehabilitation**, and a lifelong vitamin B12 plan after gastrectomy.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Durvalumab + FLOT when PD-L1 is below 1 or histology is diffuse.
- Four or six months of adjuvant CAPOX after D2 (LOMAC).
- Perioperative therapy or surgery first for cT2 N0.
- Laparoscopic or robotic versus open gastrectomy; subtotal versus total when the margin allows.
- Which immunotherapy, and for how long, in MSI-high disease; organ preservation.
- Older or frail patients: FLOT, a doublet, or surgery with adjuvant therapy.

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7

INFOGRAPHIC

Resectable gastric cancer

STOMACH

67%
2-year event-free survival with durvalumab + FLOT
59% with FLOT · HR 0.71

69%
3-year overall survival with durvalumab + FLOT
62% with FLOT · HR 0.78, p = 0.021

19%
pathological complete response
7% with FLOT alone

71.6%
grade 3–4 adverse events with D-FLOT
71.2% with FLOT: no added severe toxicity

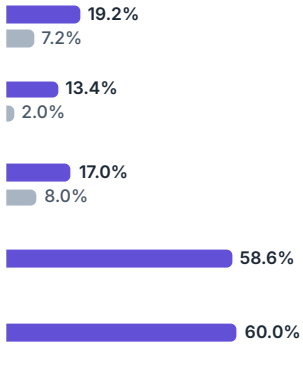
The evidence that shapes resectable gastric cancer

Trial	Design	Key result	Message
PERIOPERATIVE			
MAGIC	ECF before and after surgery vs surgery, n = 503	5-year survival 36% vs 23%, HR 0.75	Founded the perioperative strategy
FLOT4	FLOT vs ECF/ECX, n = 716	Median OS 50 vs 35 months, HR 0.77	FLOT became the backbone
MATTERHORN	Durvalumab + FLOT vs FLOT, n = 948	2-year EFS 67.4% vs 58.5%, HR 0.71; OS HR 0.78	New standard FDA Nov 2025
KEYNOTE-585	Pembrolizumab + chemotherapy vs chemotherapy, n = 804	OS 71.8 vs 55.7 months, HR 0.86; EFS HR 0.81	Negative for EFS
PRODIGY · RESOLVE	Neoadjuvant DOS; perioperative SOX (East Asia)	3-year PFS 66% vs 60%; 3-year DFS 59% vs 51%	Perioperative therapy also works after D2
ADJUVANT AFTER D2			
CLASSIC	CAPOX 6 months vs surgery, n = 1,035	5-year survival 78% vs 69%, HR 0.66	Western-compatible adjuvant standard
ACTS-GC	S-1 for 1 year vs surgery, n = 1,059	5-year survival 72% vs 61%, HR 0.67	East Asian standard, stage II
JACCRO GC-07	S-1 + docetaxel vs S-1, stage III	3-year relapse-free survival 66% vs 50%	East Asian standard, stage III
LOMAC	CAPOX 4 vs 6 months, n = 1,023	3-year DFS 69.2% vs 70.8%, non-inferior	ESMO 2026 Shorter is acceptable
RADIOTHERAPY			
INT-0116	Adjuvant chemoradiotherapy vs surgery (mostly D0–D1)	Median OS 36 vs 27 months	Compensates for inadequate surgery
ARTIST 2 · CRITICS	Adding radiotherapy after D2, or after preoperative chemotherapy	No disease-free or overall survival gain	No radiotherapy after an R0 D2

East and West: two routes to the same goal

	Europe and the Americas	East Asia
Sequence	Perioperative: chemotherapy, surgery, chemotherapy	Surgery first, then adjuvant; perioperative for bulky disease
Regimen	FLOT ± durvalumab	S-1, CAPOX/SOX, S-1 + docetaxel
Surgery	D2 in high-volume centers	D2 is routine; high laparoscopic volume

Pathological complete response (%)



A higher response rate predicted survival only in MATTERHORN. MSI-high tumors (about 8%) respond to immunotherapy and gain little from chemotherapy.

ESMO 2026

- Perioperative FLOT for ≥cT2 or cN+ (I, A); **durvalumab + FLOT preferred** where available (I, A).
- D2 gastrectomy in high-volume centers (II, A); laparoscopy is an option in expert hands (II, B).
- **LOMAC**: 4 months of adjuvant XELOX non-inferior to 6, with grade 3 or higher neuropathy 0.4% vs 5.9% and completion 84% vs 60%.

ASCO 2025–26

- MATTERHORN event-free survival (ASCO 2025) led to approval FDA 25 Nov 2025.
- KEYNOTE-585 MSI-high subgroup (ASCO 2026).
- NCCN v4.2026: universal PD-L1 testing at diagnosis; perioperative immunotherapy options for MSI-high.

KEY MESSAGE | Cure in gastric cancer needs three things done well: **a staged abdomen** (laparoscopy), **a D2 operation**, and **systemic therapy around it**, now with durvalumab for most patients.

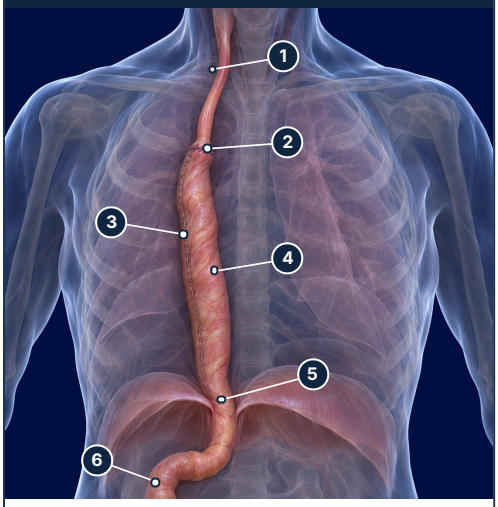
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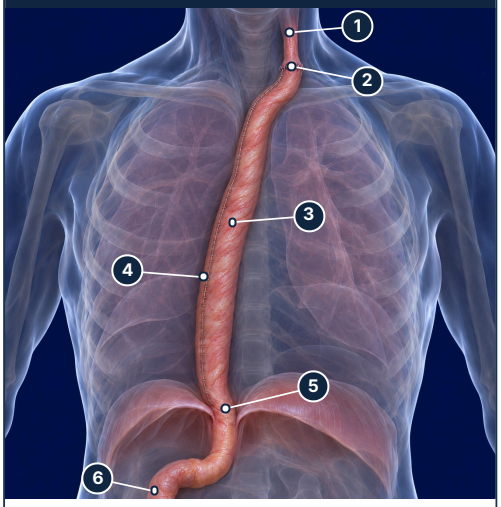
Surgical atlas: esophagectomy

3D Ivor Lewis: anastomosis in the chest



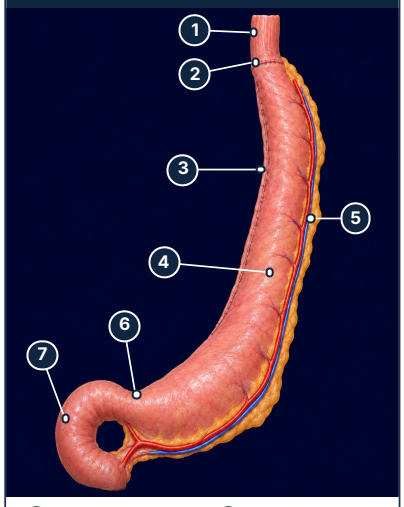
1. Esophagus
2. **Chest anastomosis**
3. Staple line
4. **Gastric conduit**
5. Hiatus
6. Pylorus, duodenum

3D McKeown or transhiatal: in the neck



1. Cervical esophagus
2. **Neck anastomosis**
3. **Gastric conduit**
4. Staple line
5. Hiatus
6. Pylorus, duodenum

3D The gastric conduit



1. Esophagus
2. Anastomosis
3. Staple line
4. **Conduit**
5. **Gastroepiploic arcade**
6. Pylorus
7. Duodenum

Choosing the approach

	Ivor Lewis	McKeown	Transhiatal
Incisions · anastomosis	Abdomen + right chest · intrathoracic	Right chest + abdomen + left neck · cervical	Abdomen + left neck · cervical
Best for	Lower third and junction (Siewert I–II)	Upper and middle third; long Barrett segment	Distal tumors when thoracotomy is unsafe
Lymphadenectomy	Two-field (abdomen + mediastinum)	Two- or three-field	Abdominal and lower mediastinal only
Strengths	Fewer recurrent-nerve injuries and strictures	Longest proximal margin; a neck leak is less dangerous	Less pulmonary morbidity
Limits	A thoracic leak is more serious	More recurrent-nerve palsy and strictures	Blind mediastinal dissection, lower nodal yield

9% pulmonary infection after minimally invasive esophagectomy; 29% after open surgery (TIME)

36% major complications with hybrid surgery (laparoscopy + thoracotomy); 64% with open (MIRO)

59% surgery-related complications after robot-assisted esophagectomy; 80% after open (ROBOT)

≥15 lymph nodes to stage adequately (NCCN); two-field dissection as standard

Principles of esophagectomy

- **R0 resection** with en-bloc lymphadenectomy; frozen section when the margin is in doubt.
- **Gastric conduit** on the right gastroepiploic arcade is preferred; colon or jejunum when the stomach is unusable.
- **Timing:** 4–8 weeks after chemoradiotherapy, 4–6 weeks after preoperative chemotherapy.
- **Minimally invasive or robotic** approaches are preferred where expertise exists (strengthened in NCCN v1.2026).
- **Volume matters:** centers doing at least 20 resections a year (ESMO 2026); enhanced-recovery pathway.

Complications to anticipate

- **Anastomotic leak**, about 10–20%: stent, endoluminal vacuum, drainage or reoperation.
- **Pneumonia and respiratory failure**: the main cause of death.
- **Recurrent laryngeal nerve palsy**, mainly with a neck anastomosis.
- **Chyle leak; conduit necrosis** (rare, grave).
- **Late:** stricture, reflux and aspiration, dumping, delayed emptying, weight loss.

NON-NEGOTIABLE — RED FLAGS

- **Operate only in a high-volume center** with intensive care and interventional endoscopy.
- **Restage before surgery**: interval metastases cancel the operation.
- **Preserve the gastric conduit**: no gastrostomy in a surgical candidate.
- **Salvage esophagectomy** after definitive chemoradiotherapy only in expert hands.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

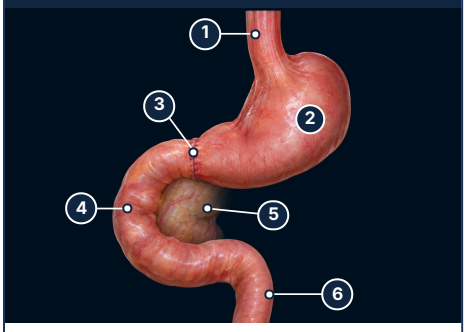
- Ivor Lewis or McKeown for mid-esophageal tumors.
- Esophagectomy or extended total gastrectomy for Siewert II.
- Three-field lymphadenectomy for upper-third squamous cancer.
- Routine feeding jejunostomy and pyloric drainage.

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Surgical atlas: gastrectomy & reconstruction

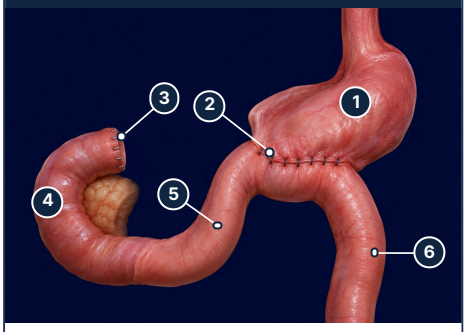
3D Distal gastrectomy: Billroth I



1. Esophagus
2. **Gastric remnant**
3. **Gastroduodenostomy**
4. Duodenum
5. Pancreas
6. Jejunum

Remnant joined to the duodenum: simple and physiological when tension-free.

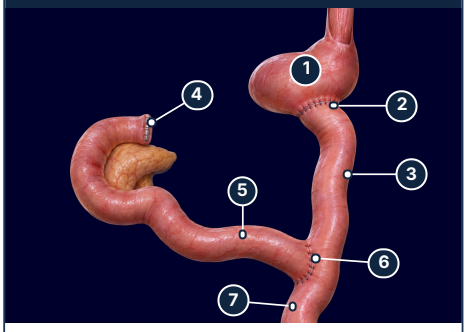
3D Distal gastrectomy: Billroth II



1. **Gastric remnant**
2. **Gastrojejunostomy**
3. Duodenal stump
4. Duodenum
5. Afferent limb
6. Efferent limb

A jejunal loop joined to the remnant; bile from the afferent limb crosses the anastomosis.

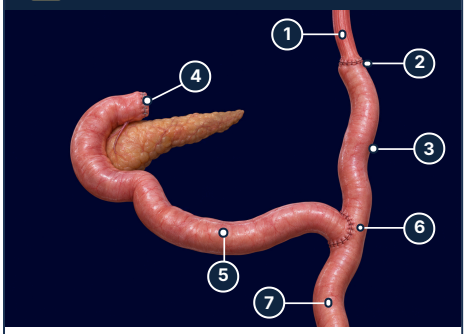
3D Distal gastrectomy: Roux-en-Y



1. **Gastric remnant**
2. **Gastrojejunostomy**
3. Roux limb
4. Duodenal stump
5. Biliopancreatic limb
6. **Jejunojejunostomy**
7. Common channel

Bile joins the food downstream: less bile reflux than Billroth II; preferred with a small remnant.

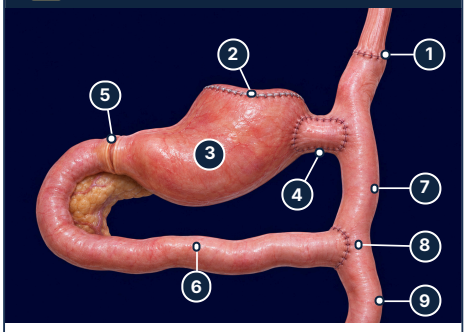
3D Total gastrectomy: Roux-en-Y



1. Esophagus
2. **Esophagojejunostomy**
3. Roux limb, 40–60 cm
4. Duodenal stump
5. Biliopancreatic limb
6. **Jejunojejunostomy**
7. Common channel

Esophagojejunostomy: for proximal or diffuse tumors and linitis.

3D Proximal gastrectomy: double tract



1. **Esophagojejunostomy**
2. Staple line
3. **Distal stomach, kept**
4. **Gastrojejunostomy**
5. Pylorus
6. Duodenal route
7. Direct jejunal route
8. **Jejunojejunostomy**
9. Common channel

Two routes for food: through the remnant and duodenum, or straight down the jejunum.

Which gastrectomy, which margin

- **Distal (subtotal)** when an adequate proximal margin is possible; **total** for proximal or diffuse tumors and linitis.
- **Margins:** 2 cm for T1; 3 cm for T2 or deeper with an expansive pattern; 5 cm for infiltrative or diffuse growth; NCCN asks for 4 cm or more.
- **Frozen section** when the margin is close or histology is diffuse.
- **Function-preserving options** for early cancer: proximal gastrectomy with double-tract reconstruction; pylorus-preserving gastrectomy.
- **At least 16 nodes** examined; 30 or more gives the most reliable stage.

Reconstruction at a glance

	Billroth I	Billroth II	Roux-en-Y	Double tract
Joined	Remnant to duodenum	Remnant to a jejunal loop	Remnant, or esophagus, to a divided jejunal limb	Esophagus and remnant to one jejunal limb
Food through the duodenum	Yes	No	No	In part
Bile meets food	In the duodenum, as before	At the gastric anastomosis	Downstream, at the jejunojejunostomy	In the duodenum and at the jejunojejunostomy
Chosen when	The duodenum reaches without tension	A Billroth I would be under tension	Small remnant; after total gastrectomy	Early upper-third cancer, to keep the distal stomach
Watch for	Tension on the anastomosis	Afferent-loop obstruction; duodenal stump leak	Roux stasis; internal hernia; duodenal stump leak	Three suture lines; the remnant still needs endoscopy

Two randomized answers on reconstruction

- **Roux-en-Y or Billroth I** after distal gastrectomy (Cochrane review 2021: 8 trials, n = 942): bile reflux is less frequent with Roux-en-Y (risk ratio 0.40, 95% CI 0.25–0.63); quality of life and weight loss do not differ; overall morbidity may be lower with Billroth I.
- **KLASS-05** (n = 138; upper-third cT1N0): proximal gastrectomy with double tract against total gastrectomy needed less vitamin B12 over 2 years (0.4 vs 2.5 mg); hemoglobin change, reflux esophagitis (2.9% in each arm) and 2-year survival (98.5% vs 100%) did not differ.

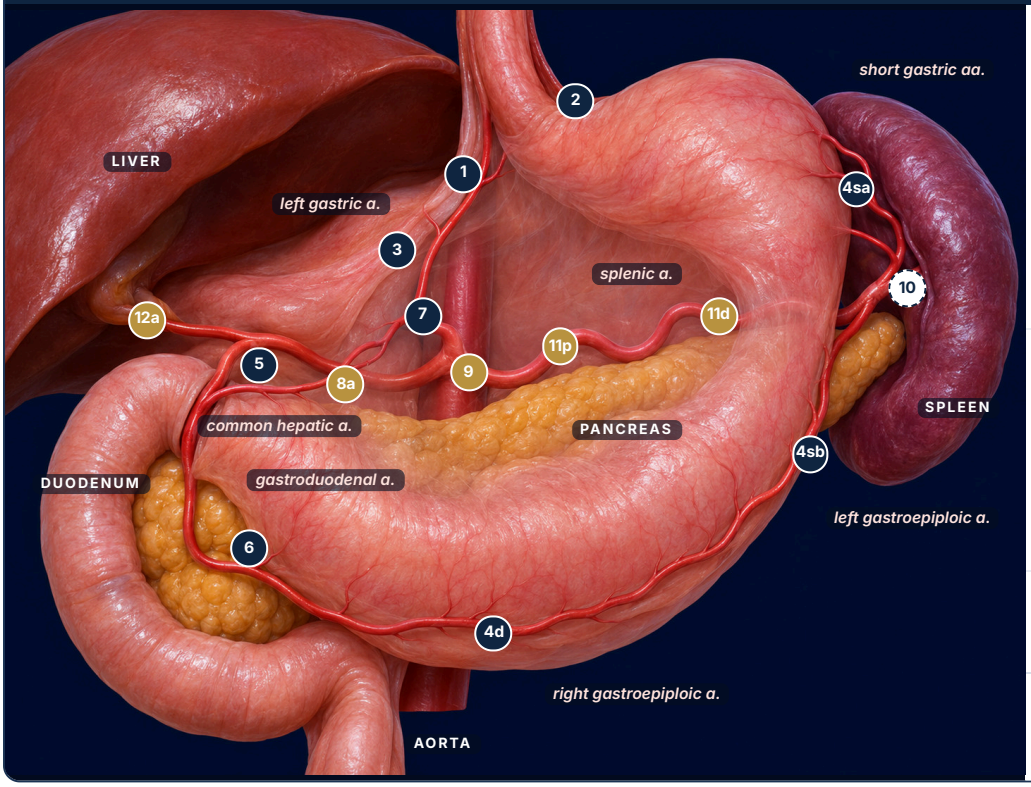
KEY MESSAGE | The **tumor** dictates the resection: site, margin and histology. The **remnant** dictates the reconstruction. Neither replaces a D2 lymphadenectomy, shown on the next page.

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Surgical atlas: D2 lymphadenectomy

3D Arterial map of the stomach with the JGCA lymph-node stations



Stations 1–7: the D1 field

- 1 Right paracardial
- 2 Left paracardial
- 3 Lesser curvature
- 4sa Short gastric vessels
- 4sb Left gastroepiploic vessels
- 4d Right gastroepiploic vessels
- 5 Suprapyloric
- 6 Infrapyloric
- 7 Left gastric artery

Along the celiac branches: added for D2

- 8a Common hepatic artery
- 9 Celiac trunk
- 11p Proximal splenic artery
- 11d Distal splenic artery
- 12a Hepatoduodenal ligament, along the hepatic artery

Only when the greater curvature is involved

- (10) Splenic hilum

The stomach is drawn translucent so that the celiac trunk and its three branches show through. Nodes are numbered by the artery they follow.

What D2 means (JGCA)

Dissection	Distal gastrectomy	Total gastrectomy
D1	1, 3, 4sb, 4d, 5, 6, 7	1–7
D1+	D1 + 8a, 9	D1 + 8a, 9, 11p
D2	D1 + 8a, 9, 11p, 12a	D1 + 8a, 9, 11p, 11d, 12a

Each artery carries its own stations

Artery	Stations it carries
Left gastric	**7** at its trunk; **1** and **3** on its branches to the cardia and lesser curvature
Common and proper hepatic	**8a**; **12a** in the hepatoduodenal ligament
Right gastric	**5**, above the pylorus; then **3**
Gastroduodenal, right gastroepiploic	**6** below the pylorus; **4d** along the greater curvature
Splenic	**11p**, **11d**; **10** at the hilum; then **4sb** and **4sa**
Celiac trunk · left inferior phrenic	**9** · **2**, to the left of the cardia

Six surgical questions answered by randomized trials

Question	Trial	Answer
D1 or D2?	Dutch D1D2, 15 years	Gastric-cancer death 37% vs 48%: D2, with spleen and pancreas preserved
Remove the spleen?	JCOG0110	No survival gain, more morbidity: preserve unless the greater curvature is invaded
Bursectomy? Para-aortic nodes?	JCOG1001 · JCOG9501	No benefit from either
Laparoscopic distal gastrectomy?	KLASS-01 · CLASS-01 · KLASS-02	Non-inferior survival in stage I and in locally advanced disease, in expert hands
Thoracotomy for Siewert II–III?	JCOG9502	No: transhiatal approach when esophageal invasion is 3 cm or less
Resect the primary in stage IV?	REGATTA · RENAISSANCE	No survival benefit

NON-NEGOTIABLE — RED FLAGS

- **D2 lymphadenectomy** by a surgeon who does it routinely.
- **Spleen- and pancreas-preserving** dissection.
- **R0 margins**, with frozen section in diffuse-type cancer.
- **No gastrectomy for metastatic disease** except to palliate bleeding or obstruction.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

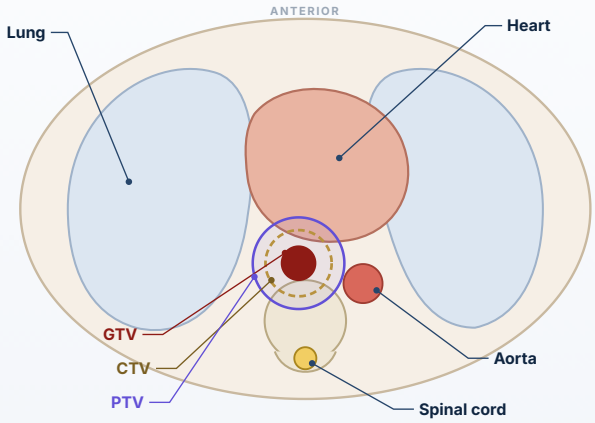
- Laparoscopic, robotic or open, according to expertise.
- Billroth I, Billroth II or Roux-en-Y after distal gastrectomy.
- Total or proximal gastrectomy for early upper-third cancer.
- Omentectomy or omentum preservation; station 10 dissection.

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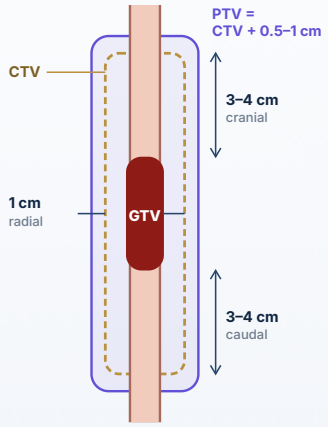
Radiotherapy planning atlas: esophagus

Axial plan at mid-thoracic level



GTV: primary and involved nodes on PET/CT, endoscopy and endoscopic ultrasound. **CTV** (gold): GTV plus margins, trimmed at bone, heart and airway. **PTV** (violet): CTV + 0.5–1 cm according to motion management.

Margins along the esophagus



CTV: 3–4 cm above and below the GTV, 1 cm radially; 1 cm around involved nodes.

Dose and fractionation

Intent	Dose	Given with	Evidence and notes
Preoperative	**41.4 Gy in 23 fractions** (CROSS); up to 50.4 Gy in 28	Weekly carboplatin–paclitaxel	Surgery about 4–8 weeks later; keep the lung dose low for the thoracotomy
Definitive	**50–50.4 Gy in 25–28 fractions**	FOLFOX, carboplatin–paclitaxel or cisplatin–5-FU	64.8, 61.6 and 66 Gy gave no gain (INT 0123, ARTDECO, CONCORDE)
Cervical esophagus	50.4 Gy; 60–66 Gy by institutional practice Negotiable	As for definitive treatment	Organ-preserving standard; surgery is not the first choice
Postoperative	45–50.4 Gy in 25–28 fractions	Fluoropyrimidine	R1 or R2 resection without preoperative radiotherapy. Palliative schedules: next page

Organ-at-risk dose limits (as in NCCN)

Organ	Planning limit
Lungs	Mean under 20 Gy; V20 20% or less; V10 40% or less; lower still before esophagectomy
Heart	Mean under 30 Gy; V30 30% or less, nearer 20% if possible
Spinal cord	Maximum 45 Gy
Liver	Mean under 25 Gy; V30 20% or less
Kidneys	Each kidney: no more than one third above 18 Gy
Stomach	Outside the target: mean under 30 Gy; spare the fundus for the conduit

Limits differ between protocols; follow the one your department has validated.

Elective nodes by tumor site

Tumor site	Elective nodal volume, if used
Cervical, upper thoracic	Supraclavicular and upper mediastinal (paratracheal)
Middle thoracic	Paraesophageal and subcarinal
Lower thoracic, junction	Lower paraesophageal, paracardial, left gastric, celiac

Simulation and delivery

- **Simulation:** supine, arms up, intravenous contrast; fast for 2–3 hours when the tumor is distal.
- **Motion and fusion:** 4D-CT for the lower third and junction; PET/CT with the endoscopic limits.
- **Technique:** IMRT or VMAT with daily image guidance.
- **Protons:** toxicity burden 2.3 times lower than IMRT, equal survival (randomized phase IIB); NRG-GI006 awaited Negotiable.

NON-NEGOTIABLE — RED FLAGS

- **Concurrent chemotherapy** with every curative course; radiotherapy alone only when chemotherapy is impossible.
- **No more than 50.4 Gy** to the thoracic esophagus outside a trial.
- **Motion management** for distal and junctional tumors.
- **Contour lungs, heart, cord, liver and kidneys** on every plan, and report the lung dose to the surgeon.
- **Fistula:** control it before any curative-dose radiotherapy, and decide at the tumor board.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- 41.4 or 50.4 Gy before surgery.
- Elective nodal irradiation or involved-field treatment.
- A boost to 60–66 Gy for cervical tumors.
- IMRT or protons.
- Induction chemotherapy before chemoradiotherapy for bulky disease.

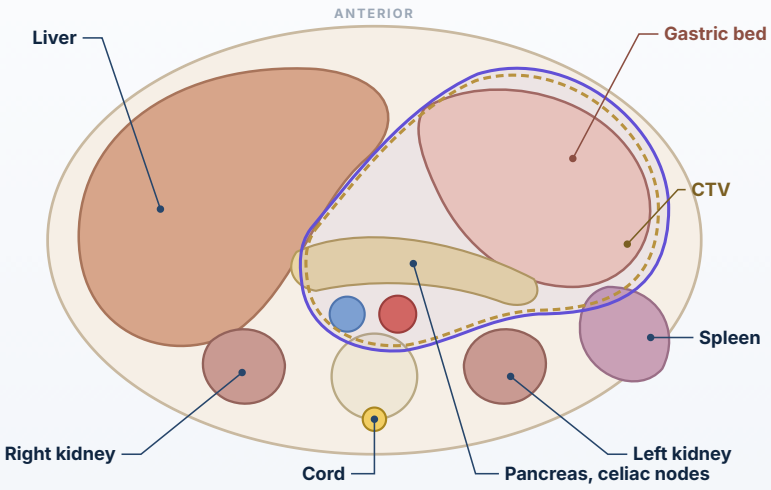
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Radiotherapy planning atlas: stomach & palliation

ESOPHAGUS STOMACH

Axial plan after gastrectomy



CTV (gold): tumor bed, anastomoses and the regional nodes for the site of the primary. **PTV** (violet): CTV + 0.5–1 cm. Both kidneys, the liver and the cord are contoured on every plan.

Where radiotherapy still has a role in gastric cancer

- **Not** after an R0 D2 gastrectomy (ARTIST 2, CRITICS), and **not** added to perioperative chemotherapy (TOPGEAR).
- **Yes** after less than a D2 dissection or an R1 resection when no preoperative therapy was given: 45 Gy in 25 fractions with a fluoropyrimidine (INT-0116).
- **Yes** for localized disease that cannot be resected or operated: 45–50.4 Gy with a fluoropyrimidine.
- **Yes** for palliation: bleeding, pain, obstruction.
- Junctional tumors unfit for FLOT: preoperative chemoradiotherapy (chapter 6).

Adjuvant volumes by site of the primary

Primary site	Tumor bed and anastomoses	Nodal volumes
Proximal third, cardia	Tumor bed, esophagojejunal anastomosis, medial left hemidiaphragm	Perigastric, celiac, suprapancreatic, splenic hilum, lower paraesophageal
Body	Tumor bed, remnant, anastomosis	Perigastric, celiac, suprapancreatic, splenic hilum, porta hepatis, pancreaticoduodenal
Antrum, pylorus	Tumor bed, duodenal stump, anastomosis	Perigastric, celiac, suprapancreatic, porta hepatis, pancreaticoduodenal; the splenic hilum can be spared

Organ-at-risk dose limits (as in NCCN)

Organ	Planning limit
Kidneys	Each kidney: no more than one third above 18 Gy; document split function if one lies in the field
Liver	Mean under 25 Gy; V30 20% or less
Spinal cord	Maximum 45 Gy
Heart	Mean under 30 Gy; V30 30% or less
Small bowel	V45 under 195 cc; no hot spot in the bowel

Planning and support

- **Before contouring:** preoperative CT, endoscopy, operation note and pathology define the tumor bed.
- **Simulation:** fasting for 2–3 hours, intravenous and oral contrast, 4D-CT; IMRT or VMAT with daily image guidance.
- **Chemotherapy:** capecitabine 825 mg/m² twice daily on treatment days, or infusional 5-FU.
- **Support:** antiemetic before each fraction, acid suppression, weekly weight and dietitian review.

Palliative radiotherapy for both cancers

Problem	Schedule	What to expect
Bleeding tumor	8 Gy once, 20 Gy in 5 or 30 Gy in 10	Bleeding stops in about three patients in four; choose the schedule by life expectancy
Dysphagia	20 Gy in 5 or 30 Gy in 10; brachytherapy 12 Gy once	Slower than a stent, but longer-lasting with fewer complications (SIREC)
Dysphagia needing relief within days	Self-expanding metal stent	Radiotherapy after the stent did not improve swallowing or survival; it reduced bleeding (ROCS)
Pain, bone or nodal metastasis	8 Gy once or 20 Gy in 5	Standard palliative practice
Oligometastatic squamous cancer	Stereotactic radiotherapy to every site, with systemic therapy Negotiable	PFS 15.3 vs 6.4 months in a randomized phase II trial (ESO-Shanghai 13)

NON-NEGOTIABLE — RED FLAGS

- **No adjuvant radiotherapy after an R0 D2 resection.**
- **Both kidneys contoured**, with renal function documented before upper-abdominal radiotherapy.
- **Antiemetic prophylaxis and nutritional support** throughout.
- **A stent, not radiotherapy,** when swallowing must be restored within days.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Chemoradiotherapy or chemotherapy alone after a D1 dissection with positive nodes.
- One, five or ten fractions for bleeding.
- Brachytherapy or external beam for dysphagia.
- Stereotactic radiotherapy for oligometastases.

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DECISION ALGORITHM

10 Advanced gastric, junctional & esophageal adenocarcinoma: first line

STOMACH / GEJ / ESOPHAGUS · ADENOCARCINOMA

Unresectable or metastatic adenocarcinoma of the stomach, junction or esophagus

HER2, MMR/MSI, PD-L1 and CLDN18.2 reported · *DPYD* genotype · performance status · nutrition and symptom plan · palliative care introduced

HER2-positive · about 15–20%

Zanidatamab + tislelizumab + chemotherapy

Preferred where available: IHC 3+ or IHC 2+ with ISH amplification, any PD-L1 (I, A)

FDA Aug 2026 · ASCO · ESMO

Zanidatamab + chemotherapy

IHC 3+ only in the US label (I, B)

Trastuzumab + chemotherapy

With **pembrolizumab when PD-L1 CPS is 1 or more** (KEYNOTE-811) (I, A); without it when below 1 (ToGA)

Mismatch-repair deficient or MSI-high · about 3–5%

Checkpoint inhibitor + fluoropyrimidine–oxaliplatin

Whatever the PD-L1 score (II, A)

Immunotherapy alone

Pembrolizumab, or nivolumab + ipilimumab: case by case, for low-burden disease or when chemotherapy is unsuitable

Negotiable

HER2-negative, mismatch-repair proficient · about 4 in 5

Fluoropyrimidine + oxaliplatin

FOLFOX or CAPOX, plus the partner chosen from **PD-L1 and CLDN18.2** in the matrix below (I, A)

Fit, with a high tumor burden

Docetaxel + FOLFOX (TFOX): survival 15.1 vs 12.7 months in GASTFOX (I, B) Negotiable

Frail or older

Oxaliplatin–capecitabine at 60% of the dose: non-inferior and better tolerated (GO2)

HER2-negative, mismatch-repair proficient: the partner for chemotherapy

	CLDN18.2 negative	CLDN18.2 positive at least 75% of cells, moderate to strong
PD-L1 CPS 1 or more	**Chemotherapy + PD-1 inhibitor** Nivolumab, pembrolizumab or tislelizumab. The benefit grows with the score and is clearest at CPS 5–10 or more.	**Either partner** Favor the PD-1 inhibitor as PD-L1 rises; favor zolbetuximab when the score is low or immunotherapy is contraindicated. Negotiable
PD-L1 CPS below 1	**Chemotherapy alone** No PD-1 inhibitor: pooled hazard ratio 0.91 (0.75–1.09) in the FDA analysis.	**Chemotherapy + zolbetuximab** SPOTLIGHT and GLOW (I, A) FDA Oct 2024

Relapse within 6 months of perioperative FLOT

Treat as refractory to oxaliplatin and taxane: FOLFIRI-based therapy (I, B), or second-line options (chapter 12)

After 4–6 months of oxaliplatin

Stop it before neuropathy disables; continue the fluoropyrimidine with the antibody. Imaging every 6–9 weeks

Europe reads PD-L1 differently

Nivolumab CPS 5 or more, pembrolizumab CPS 1 or more, tislelizumab TAP 5% or more ESMO 2026

NON-NEGOTIABLE — RED FLAGS

- **Four biomarkers before cycle 1:** HER2, MMR/MSI, PD-L1, CLDN18.2.
- **No PD-1 inhibitor when PD-L1 is below 1** in mismatch-repair proficient disease (US labels restricted in 2025).
- **No HER2-directed therapy** without IHC 3+ or IHC 2+ with ISH amplification.
- **Zolbetuximab:** full antiemetic prophylaxis and a slow first infusion; nausea and vomiting peak in cycle 1.
- **Zanidatamab:** antidiarrheal prophylaxis, and no 5-FU bolus with FOLFOX.
- ***DPYD* genotype** before any fluoropyrimidine, unless treatment cannot wait (EMA 2020; FDA boxed warning 2025–26).
- **Not outside trials:** nivolumab + ipilimumab + chemotherapy (ATTRACTION-6), anti-TIGIT combinations (STAR-221), bemarituzumab.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- PD-1 inhibitor or zolbetuximab when both markers are positive.
- A PD-1 inhibitor when CPS is 1–4: the gain is small.
- Zanidatamab with or without tislelizumab when PD-L1 is below 1.
- Triplet chemotherapy for fit patients with a high tumor burden.
- Immunotherapy alone or with chemotherapy in MSI-high disease.
- When to stop oxaliplatin, and what to maintain.

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INFOGRAPHIC

10 Advanced gastric, junctional & esophageal adenocarcinoma: first line

STOMACH / GEJ / ESOPHAGUS · ADENOCARCINOMA

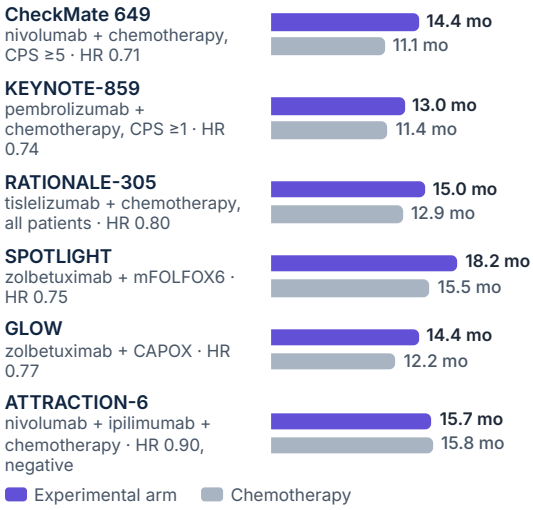
27.0 mo
median survival with zanidatamab + tislelizumab + chemotherapy
19.6 months with trastuzumab + chemotherapy · HERIZON-GEA-01, 35-month update

0.71
hazard ratio for death with nivolumab + chemotherapy when CPS is 5 or more
14.4 vs 11.1 months · CheckMate 649

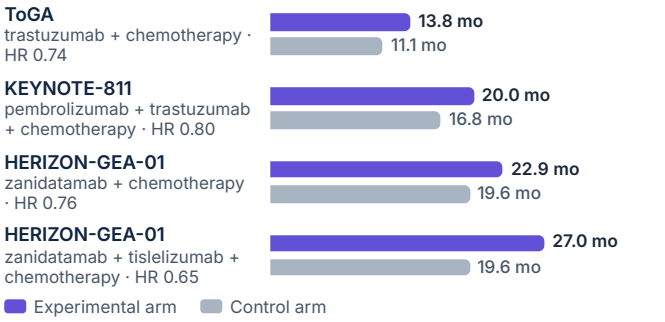
18.2 mo
median survival with zolbetuximab + mFOLFOX6
15.5 months with placebo · SPOTLIGHT

0.91
hazard ratio with a PD-1 inhibitor when PD-L1 is below 1
95% CI 0.75–1.09 · FDA pooled analysis

HER2-negative: median overall survival (months)



HER2-positive: median overall survival (months)



How often is each marker positive?

Biomarker	Positive in about	Opens the door to
HER2	15–20%	Zanidatamab or trastuzumab; later T-DXd
CLDN18.2 (≥75% of cells)	38%	Zolbetuximab
PD-L1 CPS ≥1 / ≥5	80% / 60%	PD-1 inhibitor
dMMR or MSI-high	3–5%	Immunotherapy-based treatment

Class effects to anticipate

- **Zolbetuximab:** nausea and vomiting, mostly during the first infusions; hypoalbuminemia.
- **Zanidatamab:** diarrhea, infusion reactions, fall in ejection fraction.
- **PD-1 inhibitors:** immune-related events in any organ, including months later.
- **Trastuzumab:** cardiac monitoring every 3 months.
- **Oxaliplatin:** cumulative neuropathy: stop in time.

What did not work, and what it teaches

Trial	Tested	Result	Lesson
ATTRACTION-6	Nivolumab + ipilimumab + chemotherapy vs chemotherapy	OS 15.7 vs 15.8 months, HR 0.90	More toxicity, no gain ASCO 2026
STAR-221	Domvanalimab + zimberelimab + chemotherapy vs nivolumab + chemotherapy	No survival improvement	Anti-TIGIT adds nothing ESMO GI 2026
FORTITUDE-101	Bemarituzumab + mFOLFOX6, FGFR2b-positive	HR 0.61 early, 0.82 with longer follow-up	Benefit faded; program stopped
JACOB	Pertuzumab added to trastuzumab + chemotherapy	OS 17.5 vs 14.2 months, not significant	Dual antibody blockade failed
KEYNOTE-062	Pembrolizumab + chemotherapy, CPS ≥1	OS 12.5 vs 11.1 months, not significant	Selection by PD-L1 matters

ASCO 2026 Guideline update and data

- **Guideline update (Feb 2026):** test HER2, MMR/MSI, PD-L1 and CLDN18.2; choose the partner from PD-L1 (1 or more) and CLDN18.2.
- **Living update (Jul 2026):** HER2-positive disease: HER2 antibody + doublet, with immunotherapy when PD-L1 is 1 or more.
- **HERIZON-GEA-01** (N Engl J Med 2026): PFS 12.4 vs 8.1 months, HR 0.63–0.65 FDA 25 Aug 2026.
- **ATTRACTION-6** negative.

ESMO 2026 Guideline and Madrid

- Zanidatamab + tislelizumab + chemotherapy for HER2-positive disease where available (I, A).
- Zolbetuximab (I, A) (MCBS 2); TFOX for selected fit patients (I, B).
- **HERIZON update:** OS 22.9 vs 19.6 months with zanidatamab + chemotherapy (HR 0.76); 27.0 with the triplet (HR 0.65).
- **FORTITUDE-101/-102:** FGFR2-amplified subgroups, exploratory.

KEY MESSAGE | First-line therapy is now chosen by **four biomarkers**, not by organ: HER2, mismatch repair, PD-L1 and CLDN18.2. Chemotherapy alone is the right answer only when all four are negative.

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DECISION ALGORITHM

11 Advanced esophageal squamous cell carcinoma

ESOPHAGUS · SQUAMOUS

Unresectable, recurrent or metastatic squamous cell carcinoma of the esophagus

PD-L1 reported (CPS, TPS or TAP) · mismatch repair · swallowing and nutrition plan · airway and fistula risk · performance status

PD-L1 positive: 1 or more

Platinum doublet + PD-1 inhibitor

- **Pembrolizumab** + cisplatin–5-FU (KEYNOTE-590)
- **Nivolumab** + cisplatin–5-FU (CheckMate 648)
- **Tislelizumab** + platinum with a fluoropyrimidine or paclitaxel (RATIONALE-306)
- Toripalimab, camrelizumab or sintilimab + paclitaxel–cisplatin Regional

Chemotherapy-free: nivolumab + ipilimumab

Tumor-cell PD-L1 of 1% or more (I, B). More early progression than with chemotherapy: not for bulky, symptomatic disease Negotiable

Duration

Platinum for up to 6 cycles; immunotherapy to progression or 2 years

PD-L1 below 1

Chemotherapy alone

A fluoropyrimidine with cisplatin or oxaliplatin, or carboplatin–paclitaxel

Why no PD-1 inhibitor

Pooled hazard ratio for death 1.10 (0.76–1.58) in the FDA analysis; US labels restricted in 2025

Europe reads PD-L1 differently

Nivolumab TPS 1% or more; pembrolizumab CPS 10 or more; tislelizumab TAP 5% or more; toripalimab CPS 1 or more ESMO 2026

Local problems first

Dysphagia

A stent restores swallowing within days; external beam or brachytherapy lasts longer (chapter 9)

Airway invasion or fistula

Covered stent; tumor response can open a fistula: warn and plan

Oligometastatic disease

Systemic therapy first; local therapy to every site in responders: PFS 15.3 vs 6.4 months (ESO-Shanghai 13) Negotiable

Progression

Reassess fitness, nutrition and goals of care before every new line

No previous immunotherapy

PD-1 inhibitor alone

Nivolumab (ATTRACTION-3) or tislelizumab (RATIONALE-302); pembrolizumab when CPS is 10 or more (KEYNOTE-181)

After immunotherapy

Taxane or irinotecan

Paclitaxel, docetaxel or irinotecan; clinical trial whenever possible

New in 2026

Izalontamab brengitecan

EGFR × HER3 antibody–drug conjugate after platinum and PD-1 therapy: OS 9.8 vs 7.2 months, HR 0.63 (PANKU-Esophagus01); approved in China ASCO 2026 Regional

Choosing the chemotherapy backbone

- **Cisplatin–5-FU:** the partner in KEYNOTE-590 and CheckMate 648.
- **Paclitaxel–platinum:** the partner in the East Asian trials and an option in RATIONALE-306.
- **Oxaliplatin or carboplatin** instead of cisplatin when renal function, hearing or frailty require it.

Squamous cancer is not adenocarcinoma

- No HER2 or CLDN18.2 strategy; zolbetuximab and trastuzumab have no role.
- Malnutrition and aspiration are the rule: feeding route decided on day 1.
- Second primaries of the head and neck and lung share the same carcinogens.

NON-NEGOTIABLE — RED FLAGS

- **PD-L1 before first-line therapy**, with the assay and score named in the report.
- **No PD-1 inhibitor when PD-L1 is below 1** (US labels, ASCO): chemotherapy alone.
- **Swallowing, weight and feeding route** reviewed at every visit.
- **Fistula risk** assessed before treatment when the airway is invaded.
- **No second PD-1 inhibitor** after progression on one, outside a trial.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

- Which PD-1 inhibitor, and which chemotherapy partner.
- Nivolumab + ipilimumab versus chemotherapy with immunotherapy.
- A PD-1 inhibitor when PD-L1 is 1–9: the gain is smaller.
- Local therapy for oligometastatic disease.
- Stopping immunotherapy at 2 years in responders.

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11 INFOGRAPHIC

Advanced esophageal squamous cell carcinoma

ESOPHAGUS · SQUAMOUS

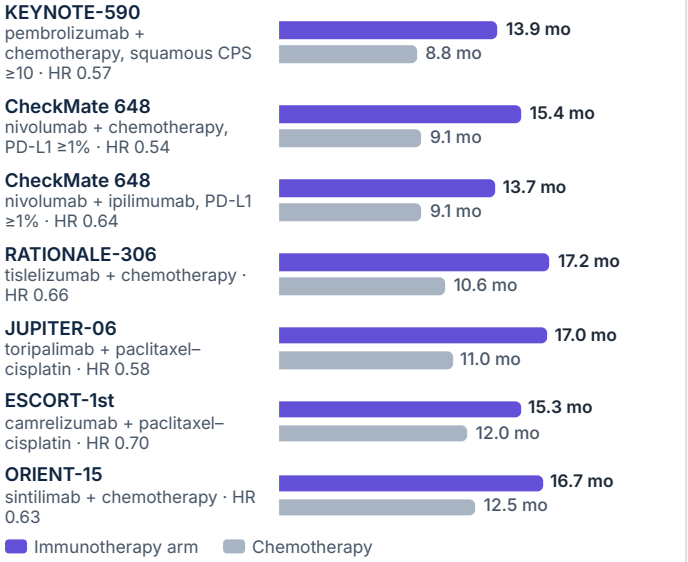
17.2 mo
median survival with tislelizumab + chemotherapy
10.6 months with chemotherapy · RATIONALE-306

0.57
hazard ratio for death with pembrolizumab + chemotherapy, squamous CPS 10 or more
13.9 vs 8.8 months · KEYNOTE-590

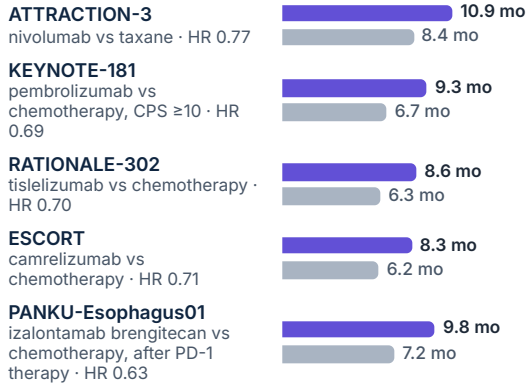
1.10
hazard ratio with a PD-1 inhibitor when PD-L1 is below 1
95% CI 0.76–1.58 · FDA pooled analysis

9.8 mo
median survival with izalontamab brengitecan after PD-1 therapy
7.2 months with chemotherapy · PANKU-Esophagus01

First line: median overall survival (months)



Second line: median overall survival (months)



Reading the first-line trials

- These six trials enrolled patients whatever the PD-L1 score; the benefit rises with PD-L1.
- CPS, TPS and TAP count different cells: TPS counts tumor cells only; NCCN accepts TAP in place of CPS.
- Most patients in the East Asian trials received paclitaxel–cisplatin.

The same drug, two thresholds: the label map

First-line regimen	United States: FDA label, NCCN	Europe: ESMO 2026
Pembrolizumab + platinum–fluoropyrimidine	PD-L1 CPS 1 or more	CPS 10 or more
Nivolumab + platinum–fluoropyrimidine	PD-L1 1 or more	Tumor-cell PD-L1 1% or more
Nivolumab + ipilimumab	PD-L1 1 or more	Tumor-cell PD-L1 1% or more
Tislelizumab + platinum-based chemotherapy	PD-L1 1 or more	TAP 5% or more
Toripalimab + paclitaxel–cisplatin	Not licensed for this use	CPS 1 or more (EU label: any PD-L1)

ASCO 2026 Guideline and data

- **Guideline update:** PD-L1 of 1 or more: immunotherapy with chemotherapy, or nivolumab + ipilimumab. Below 1: chemotherapy.
- **Second line without previous immunotherapy:** nivolumab or tislelizumab; pembrolizumab when CPS is 10 or more.
- **PANKU-Esophagus01** (n = 497, China; JAMA 2026): OS 9.8 vs 7.2 months, HR 0.63; PFS 4.2 vs 2.0 months, HR 0.50; grade 3 or higher treatment-related events 85% vs 59%.
- **NCCN v4.2026:** add a PD-1 inhibitor when CPS is 1 or more; TAP is interchangeable with CPS.

ESMO 2026 Guideline

- First line, with chemotherapy: nivolumab TPS 1% or more, pembrolizumab CPS 10 or more, tislelizumab TAP 5% or more, toripalimab CPS 1 or more (I, A).
- Nivolumab + ipilimumab when tumor-cell PD-L1 is 1% or more (I, B).
- Second line: nivolumab or tislelizumab if no previous immunotherapy; otherwise a taxane or irinotecan.
- **KEYNOTE-975:** no role for pembrolizumab with definitive chemoradiotherapy (chapter 5).

KEY MESSAGE | In advanced squamous cancer one number decides the first line: **PD-L1**. Positive: a PD-1 inhibitor with chemotherapy. Negative: chemotherapy alone. Check which score your label uses.

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DECISION ALGORITHM

12 Later lines & special situations

ESOPHAGUS · STOMACH

Adenocarcinoma progressing after first-line therapy

Fitness and nutrition reassessed · **new biopsy for HER2** if it was positive · MMR and CLDN18.2 if never tested · goals of care discussed

HER2-positive on a new biopsy

Trastuzumab deruxtecan

6.4 mg/kg every 3 weeks. OS 14.7 vs 11.4 months against ramucirumab–paclitaxel (DESTINY-Gastric04) I, A

HER2 lost: about 3 in 10

Treat as HER2-negative

HER2-negative, fit

Ramucirumab + paclitaxel

OS 9.6 vs 7.4 months (RAINBOW) I, A

Taxane already given with FLOT

FOLFIRI ± ramucirumab: PFS 5.5 vs 3.7 months, survival not superior (RAMIRIS) ESMO 2026 Negotiable

Less fit

Ramucirumab alone, irinotecan, docetaxel or weekly paclitaxel

MSI-high, no previous immunotherapy

PD-1 inhibitor

Pembrolizumab II, A

Squamous cancer

Second-line options: chapter 11

Third line and beyond

A minority of patients reach it: choose while the patient is still fit

Any biomarker

Trifluridine–tipiracil

OS 5.7 vs 3.6 months (TAGS) I, A

Irinotecan or a taxane

If not used earlier

HER2-positive

Trastuzumab deruxtecan

If not yet given (DESTINY-Gastric01)

Anbenitamab + chemotherapy

After trastuzumab: OS 19.6 vs 9.6 months (KC-WISE) ESMO 2026 Regional

CLDN18.2-positive

Antibody–drug conjugates

Arcotatug tavatecan: OS 10.2 vs 6.9 months (G-HOPE-001). Sonesitatug vedotin: 10.0 vs 5.0 months in third line or later (CLARITY-Gastric01). Check local approval; trial or access program ESMO 2026

SPECIAL SITUATIONS

Peritoneal metastases

Systemic therapy. Cytoreduction with HIPEC did not prolong survival (GASTRIPEC-I: 14.9 vs 14.9 months; PERISCOPE II: 15.7 vs 16.6 months, with more harm). Drain symptomatic ascites.

Limited metastatic disease

Resecting the primary or the metastases gave no survival gain (REGATTA, RENAISSANCE). Exceptional responders: tumor board Negotiable.

Gastric outlet obstruction

Stent when life expectancy is short; surgical or endoscopic gastrojejunostomy when it exceeds 2–3 months.

Bleeding or dysphagia

Endoscopic control, then radiotherapy from 8 Gy once to 30 Gy in 10; stent or brachytherapy for dysphagia (chapter 9).

Venous thromboembolism

Stomach cancer scores 2 points on the Khorana scale. Consider prophylaxis in high-risk outpatients; with the primary in place, weigh the bleeding risk and prefer low-molecular-weight heparin.

Weight loss and cachexia

Screen at every visit; dietitian from day 1; enteral before parenteral feeding. Treat nausea, pain and early satiety before blaming the cancer.

NON-NEGOTIABLE — RED FLAGS

- **New biopsy before trastuzumab deruxtecan** when trastuzumab has been given.
- **Interstitial lung disease:** baseline CT, symptoms at every cycle; hold at grade 1, stop for good at grade 2 or more.
- **Ramucirumab:** not with active bleeding, uncontrolled hypertension, a recent arterial event or an unhealed wound.
- **No trastuzumab beyond progression** (T-ACT) and **no gastrectomy for survival** in metastatic disease.
- **No HIPEC outside trials.**
- **Palliative care, nutrition and goals of care** revisited at each progression.

NEGOTIABLE — INDIVIDUALIZE AT THE MDT

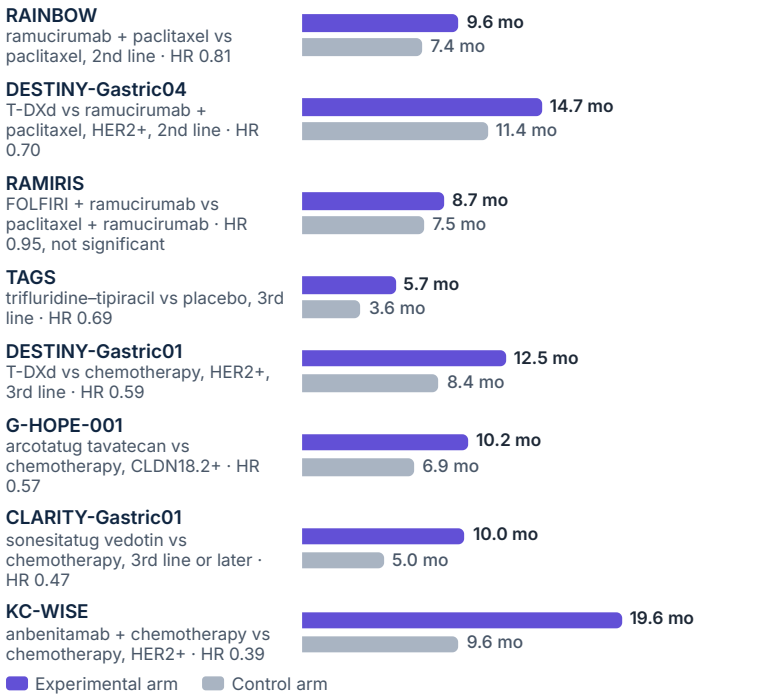
- FOLFIRI-based therapy or paclitaxel–ramucirumab after a taxane.
- Trifluridine–tipiracil or irinotecan in third line.
- Conversion surgery after an exceptional response.
- Local ablative therapy for oligoprogression.
- Oxaliplatin rechallenge after a long interval.
- Stopping anticancer treatment: best supportive care is a valid line.

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Later lines, special situations & survivorship

ESOPHAGUS · STOMACH

Later lines: median overall survival (months)



Trastuzumab deruxtecan: safe use

- HER2 confirmed on a biopsy taken after trastuzumab.
- Three-drug antiemetic prophylaxis.
- Interstitial lung disease in about 10–14%, mostly low grade: ask about cough and breathlessness at every cycle.
- Blood count before each dose; ejection fraction at baseline and when clinically indicated.

Ramucirumab: before each cycle

- Blood pressure and urine protein.
- Bleeding, thrombosis, wound healing: stop 4 weeks before planned surgery.
- Neutropenia with paclitaxel.

SURVIVORSHIP AFTER CURATIVE TREATMENT

Follow-up

	ESMO 2026	NCCN, in summary
Clinic visits	Every 3 months for 2 years, then every 6 months to at least 5 years (IV, B)	Every 3–6 months for 1–2 years, every 6–12 months to year 5, then yearly
Imaging, endoscopy	Guided by symptoms and by what salvage is still possible	By stage and treatment received; closer after organ-preserving treatment
Blood tests	Vitamin B12, iron, folate, zinc, vitamin D; bone density	Blood count and chemistry; B12 and iron after gastrectomy

Survivorship checklist

- Stop smoking and alcohol: second cancers follow squamous carcinoma.
- Eradicate *H. pylori* after endoscopic resection or partial gastrectomy.
- Genetic counseling when criteria are met (chapter 1).
- Screen for distress; plan the return to work.
- Vaccinate after splenectomy.

Life after gastrectomy or esophagectomy

Problem	Why	What to do
Weight loss, early satiety	Small reservoir, less appetite	Six small meals, dietitian, oral supplements; try pancreatic enzymes
Dumping	Rapid emptying: early within 30 minutes; late at 1–3 hours with hypoglycemia	Fewer simple sugars, fluids between meals, more protein and fiber; acarbose or octreotide if refractory
Vitamin B12 deficiency	No intrinsic factor after total gastrectomy	B12 1 mg intramuscularly every 3 months, or high-dose oral B12 daily, for life
Iron deficiency, bone loss	Bypassed duodenum, less acid, poor calcium and vitamin D uptake	Iron, intravenous if needed; calcium and vitamin D; bone density scan
Reflux, stricture, slow emptying	Vagotomy, anastomosis, conduit	Upright after meals, acid suppression, dilation, prokinetics
Diarrhea, fat malabsorption	Exocrine insufficiency, bacterial overgrowth	Pancreatic enzymes; antibiotics when overgrowth is confirmed

ASCO 2025–26

- **DESTINY-Gastric04** (ASCO 2025): T-DXd is the second-line standard for HER2-positive disease; quality-of-life data followed at ASCO 2026.
- **ASCO guideline:** ramucirumab + paclitaxel second line; ramucirumab + FOLFIRI is an option.

ESMO 2026

- **Madrid:** G-HOPE-001, CLARITY-Gastric01 and KC-WISE positive; RAMIRIS not superior for survival.
- **Guideline:** ramucirumab + paclitaxel (I, A), T-DXd (I, A), trifluridine–tipiracil (I, A); no surgery for metastatic disease.

KEY MESSAGE | After first line, **re-test before you re-treat**, and treat the patient as much as the tumor: nutrition, symptoms and an honest conversation about goals belong to every line.

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MASTER CHECKLIST

The non-negotiables: one-page checklist

ESOPHAGUS STOMACH

Thirty-two steps that should never be traded away. Read them aloud at the tumor board: each one protects cure, safety, or the validity of every decision that follows.

Diagnosis and biomarkers chapters 1–2

- [ ] Histology named: squamous cell carcinoma or adenocarcinoma.
- [ ] **HER2, MMR/MSI, PD-L1 and CLDN18.2** for every advanced adenocarcinoma; PD-L1 and MMR for squamous cancer.
- [ ] **MMR/MSI before perioperative chemotherapy.**
- [ ] *DPYD* genotype before any fluoropyrimidine.
- [ ] Nutritional assessment and a feeding plan at the first visit.

Staging chapter 3

- [ ] AJCC 8th edition, with the 2-cm rule and the Siewert type at the junction.
- [ ] PET/CT before esophagectomy or definitive chemoradiotherapy.
- [ ] **Staging laparoscopy with washings** before treating resectable gastric cancer.
- [ ] Bronchoscopy for tumors at or above the carina.
- [ ] Tumor-board decision before the first treatment.

Early disease chapter 4

- [ ] **Resect a visible lesion; never ablate it first.**
- [ ] En-bloc specimen and expert pathology; a non-curative resection goes back to the tumor board.
- [ ] Eradicate residual Barrett epithelium; eradicate *H. pylori*.

Localized disease chapters 5–7

- [ ] **Squamous cancer:** chemoradiotherapy, then surgery if fit. No dose above 50.4 Gy. No pembrolizumab with chemoradiotherapy.
- [ ] **Adenocarcinoma, fit patient:** perioperative FLOT, with durvalumab for junctional and gastric tumors. No routine radiotherapy.
- [ ] Adjuvant nivolumab only for residual disease after chemoradiotherapy and R0 resection.
- [ ] MSI-high tumors: never chemotherapy alone by default.

Surgery chapter 8

- [ ] High-volume center; R0 resection.
- [ ] **Esophagectomy:** 15 nodes or more. **Gastrectomy:** D2 with 16 nodes or more, spleen and pancreas preserved.
- [ ] No gastrostomy in a candidate for esophagectomy.
- [ ] No resection for survival in metastatic disease; no HIPEC outside trials.

Radiotherapy chapter 9

- [ ] Concurrent chemotherapy with every curative course.
- [ ] Motion management for distal tumors; lungs, heart, cord, liver and kidneys contoured and reported.
- [ ] **No adjuvant radiotherapy after an R0 D2 gastrectomy.**
- [ ] A stent, not radiotherapy, when swallowing must return within days.

Advanced disease chapters 10–12

- [ ] **No PD-1 inhibitor when PD-L1 is below 1** in mismatch-repair proficient disease.
- [ ] HER2-directed therapy only for IHC 3+ or IHC 2+ with ISH amplification; **new biopsy before trastuzumab deruxtecan**.
- [ ] Zolbetuximab and trastuzumab deruxtecan: full antiemetic prophylaxis. Trastuzumab deruxtecan: watch for interstitial lung disease.
- [ ] Palliative care and nutrition from the first line; goals of care at every progression.

Follow-up and survivorship chapter 12

- [ ] Vitamin B12 for life after total gastrectomy; iron, vitamin D and bone density.
- [ ] Smoking and alcohol cessation; think of a second primary.
- [ ] Genetic counseling when criteria are met; distress screening.

NEGOTIABLE — THE EIGHT QUESTIONS THE TUMOR BOARD IS THERE TO ANSWER

- Surgery or surveillance after a clinical complete response in squamous cancer.
- Durvalumab + FLOT when PD-L1 is below 1 or histology is diffuse.
- Esophagectomy or extended gastrectomy for Siewert II tumors.
- Four or six months of adjuvant CAPOX after D2.
- PD-1 inhibitor or zolbetuximab when PD-L1 and CLDN18.2 are both positive.
- Nivolumab + ipilimumab or chemotherapy with a PD-1 inhibitor in squamous cancer.
- Organ preservation after a complete response in MSI-high disease.
- Local therapy for oligometastatic disease.

HOW TO USE | A red flag is not a preference. If one of these steps cannot be done, the reason belongs in the notes, and the patient should know.

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T

LANDMARK TRIALS

Thirty-four years of evidence: the timeline

ESOPHAGUS STOMACH

The trials that changed practice, in the order they were reported. Gold dots mark the turning points; the full numbers are in the compendium that follows.

Localized esophageal cancer

1992 — RTOG 85-01
Chemoradiotherapy cures some patients without surgery.

2002 — INT 0123
50.4 Gy is enough: more dose, no gain.

2012 — CROSS
Chemoradiotherapy before surgery doubles median survival.

2012 — TIME
Minimally invasive esophagectomy cuts lung infections.

2018 — NEOCRTEC5010
The same benefit in squamous cancer.

2021 — CheckMate 577
First adjuvant immunotherapy
FDA May 2021

2021 — ARTDECO
Dose escalation fails again.

2024 — JCOG1109 NExT
Triplet DCF beats the doublet before surgery.

2025 — ESOPEC
Perioperative FLOT beats CROSS in adenocarcinoma.

2025 — SANO
Surveillance after a clinical complete response is non-inferior at 2 years.

2026 — KEYNOTE-975
Pembrolizumab adds nothing to definitive chemoradiotherapy
ESMO 2026

Resectable gastric and junctional cancer

2001 — INT-0116
Adjuvant chemoradiotherapy helps after limited surgery.

2006 — MAGIC
Perioperative chemotherapy becomes a standard.

2007 — ACTS-GC
Adjuvant S-1 after D2 gastrectomy.

2010 — Dutch D1D2
At 15 years, D2 lowers deaths from gastric cancer.

2012 — CLASSIC
Adjuvant CAPOX after D2 gastrectomy.

2019 — FLOT4
FLOT replaces ECF.

2019 — KLASS-01 · CLASS-01
Laparoscopic gastrectomy is safe in expert hands.

2021 — ARTIST 2
No radiotherapy after a D2 resection.

2023 — NEONIPIGA
MSI-high tumors respond to immunotherapy, not chemotherapy.

2024 — TOPGEAR
Preoperative chemoradiotherapy adds nothing to chemotherapy.

2025 — MATTERHORN
Durvalumab + FLOT: first perioperative immunotherapy
FDA Nov 2025

2026 — LOMAC
Four months of adjuvant CAPOX are enough
ESMO 2026

Advanced disease

2010 — ToGA
Trastuzumab for HER2-positive disease
FDA Oct 2010

2014 — RAINBOW · REGARD
Ramucirumab in second line.

2018 — TAGS
Trifluridine–tipiracil in third line.

2020 — DESTINY-Gastric01
Trastuzumab deruxtecan after trastuzumab.

2021 — CheckMate 649 · KEYNOTE-590
Immunotherapy enters first line.

2022 — CheckMate 648
PD-1 inhibition, with or without chemotherapy, in squamous cancer.

2023 — SPOTLIGHT · GLOW
CLDN18.2 becomes a target
FDA Oct 2024

2024 — FDA advisory committee
No benefit from PD-1 inhibitors when PD-L1 is below 1; labels changed in 2025.

2025 — DESTINY-Gastric04
Trastuzumab deruxtecan beats ramucirumab–paclitaxel in second line.

2026 — HERIZON-GEA-01
Zanidatamab replaces trastuzumab in first line
FDA Aug 2026

2026 — G-HOPE-001 · CLARITY-Gastric01
CLDN18.2 antibody–drug conjugates prolong survival in later lines ESMO 2026

2026 — PANKU-Esophagus01
First antibody–drug conjugate to prolong survival in squamous cancer
ASCO 2026

2012
chemoradiotherapy before esophagectomy
CROSS

2019
FLOT around gastrectomy
FLOT4

2021
immunotherapy in first line and after surgery
CheckMate 649, KEYNOTE-590, CheckMate 577

2025
immunotherapy around surgery
MATTERHORN

2026
a new HER2 standard
HERIZON-GEA-01

KEY MESSAGE | More than thirty years of trials point the same way: **treat the systemic disease early**, **operate well and only when it can cure**, and **choose drugs by biomarker**.

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Localized esophageal & junctional cancer

Trial	Year	Design	Population and arms	Key result	Status
CHEMORADIOTHERAPY AND SURGERY					
RTOG 85-01	1992	III, n = 123	Cisplatin–5-FU + 50 Gy vs radiotherapy alone, 64 Gy	5-year OS 26% vs 0%	Founded definitive chemoradiotherapy
INT 0123	2002	III, n = 218	64.8 vs 50.4 Gy with cisplatin–5-FU	2-year OS 31% vs 40%	Negative
Stahl et al.	2005	III, n = 172	Chemoradiotherapy ± surgery, squamous	OS equivalent; 2-year local control 64% vs 41%	Surgery adds local control
FFCD 9102	2007	III, n = 259	Surgery vs further chemoradiotherapy in responders	2-year OS 34% vs 40%; local control 66% vs 57%	Same message
CROSS	2012	III, n = 366	Carboplatin–paclitaxel + 41.4 Gy, then surgery vs surgery	OS 49.4 vs 24.0 months, HR 0.66; 10-year OS 38% vs 25%	Standard Cat 1
PRODIGE 5	2014	II/III, n = 267	FOLFOX vs cisplatin–5-FU with 50 Gy	PFS 9.7 vs 9.4 months	FOLFOX is an option
NEOCRTEC5010	2018	III, n = 451	Vinorelbine–cisplatin + 40 Gy, then surgery vs surgery, squamous	OS 100.1 vs 66.5 months, HR 0.71	Confirms CROSS in squamous cancer
ARTDECO	2021	III, n = 260	61.6 vs 50.4 Gy	3-year local PFS 73% vs 70%	Negative
CONCORDE	2021	II/III	66 vs 50 Gy with FOLFOX	No gain in progression-free survival	Negative
JCOG1109 NExT	2024	III, n = 601	Neoadjuvant DCF vs cisplatin–5-FU vs cisplatin–5-FU + 41.4 Gy, squamous	3-year OS 72.1% vs 62.6% (HR 0.68); with radiotherapy 68.3% (HR 0.84)	DCF standard in Japan
SANO	2025	Stepped-wedge, n = 309	Active surveillance vs surgery after a clinical complete response	2-year OS 74% vs 71%, non-inferior	Negotiable
ADENOCARCINOMA: CHEMOTHERAPY OR CHEMORADIOTHERAPY					
Neo-AEGIS	2023	III, n = 377	Perioperative chemotherapy vs CROSS	3-year OS 55% vs 57%, HR 1.03	Equivalent before FLOT
TOPGEAR	2024	III, n = 574	Perioperative chemotherapy ± preoperative 45 Gy	pCR 17% vs 8%; OS 46 vs 49 months, HR 1.05	Negative
ESOPEC	2025	III, n = 438	Perioperative FLOT vs CROSS	OS 66 vs 37 months, HR 0.70; 3-year OS 57.4% vs 50.7%	FLOT preferred I, A
IMMUNOTHERAPY					
CheckMate 577	2021	III, n = 794	Adjuvant nivolumab vs placebo after chemoradiotherapy and R0 resection with residual disease	DFS 22.4 vs 11.0 months, HR 0.69; final OS 51.7 vs 35.3, HR 0.85, not significant	FDA May 2021
KEYNOTE-975	2026	III	Pembrolizumab vs placebo with definitive chemoradiotherapy	EFS 15.2 vs 14.7 months, HR 0.97; OS HR 1.00	ESMO 2026 Negative
SKYSCRAPER-07	2026	III, n = 760	Tiragolumab + atezolizumab, atezolizumab or placebo after chemoradiotherapy, squamous	PFS HR 0.82, not significant; atezolizumab alone OS HR 0.69, descriptive	Not licensed
TECHNIQUE AND EARLY DISEASE					
TIME	2012	RCT, n = 115	Minimally invasive vs open esophagectomy	Pulmonary infection 9% vs 29%	Minimally invasive preferred
MIRO	2019	III, n = 207	Hybrid vs open esophagectomy	Major complications 36% vs 64%	Hybrid approach valid
ROBOT	2019	RCT, n = 112	Robot-assisted vs open esophagectomy	Complications 59% vs 80%	Robotic approach valid
JCOG0508	2019	Single-arm, n = 176	Endoscopic resection + selective chemoradiotherapy, cT1b squamous	3-year OS 92.6%	Organ preservation
MD Anderson	2020	IIB, n = 145	Proton vs intensity-modulated radiotherapy	Toxicity burden 2.3 times lower; survival equal	NRG-GI006 pending
JCOG0502	2021	Cohorts, n = 368	Esophagectomy vs chemoradiotherapy, cT1b N0 squamous	5-year OS 86.5% vs 85.5%	Both valid

READING THIS PAGE | Chemoradiotherapy before surgery for squamous cancer, FLOT around surgery for adenocarcinoma; **more radiation dose and concurrent pembrolizumab both failed.**

FDA Aug 2026 US approval, month and year · Negative primary endpoint not met · Regional data or approval limited to one region · ASCO 2026 · ESMO 2026 presented in 2026 · Year: main publication or presentation. OS overall survival; PFS progression-free survival; DFS disease-free survival; EFS event-free survival; RFS relapse-free survival; pCR pathological complete response; HR hazard ratio.

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Resectable gastric & junctional adenocarcinoma

Trial	Year	Design	Population and arms	Key result	Status
PERIOPERATIVE AND ADJUVANT CHEMOTHERAPY					
MAGIC	2006	III, n = 503	Perioperative ECF vs surgery	5-year OS 36% vs 23%, HR 0.75	Perioperative strategy
ACTS-GC	2007	III, n = 1,059	S-1 for 1 year vs surgery alone after D2	5-year OS 71.7% vs 61.1%, HR 0.67	East Asian standard
CLASSIC	2012	III, n = 1,035	CAPOX for 6 months vs surgery alone after D2	3-year DFS 74% vs 59%; 5-year OS 78% vs 69%	Adjuvant standard
FLOT4-AIO	2019	II/III, n = 716	FLOT vs ECF or ECX	OS 50 vs 35 months, HR 0.77	Backbone Cat 1
JACCRO GC-07	2019	III, n = 915	S-1 + docetaxel vs S-1, stage III	3-year RFS 66% vs 50%, HR 0.63	East Asian standard, stage III
PRODIGY	2021	III, n = 530	Neoadjuvant DOS + surgery + S-1 vs surgery + S-1	3-year PFS 66.3% vs 60.2%, HR 0.70	Regional
RESOLVE	2021	III, n = 1,022	Perioperative SOX vs adjuvant CAPOX	3-year DFS 59.4% vs 51.1%, HR 0.77	Regional
LOMAC	2026	III, n = 1,023	Adjuvant XELOX for 4 vs 6 months after D2	3-year DFS 69.2% vs 70.8%, non-inferior; neuropathy 0.4% vs 5.9%	ESMO 2026
RADIOTHERAPY					
INT-0116	2001	III, n = 556	Surgery ± 5-FU chemoradiotherapy, 45 Gy	OS 36 vs 27 months	After less than D2
CRITICS	2018	III, n = 788	Postoperative chemotherapy vs chemoradiotherapy after preoperative chemotherapy	OS 43 vs 37 months, HR 1.01	Negative
ARTIST 2	2021	III, n = 546	S-1 vs SOX vs SOX + radiotherapy, node-positive after D2	3-year DFS 64.8%, 74.3% and 72.8%	No radiotherapy after D2
PERIOPERATIVE IMMUNOTHERAPY					
NEONIPIGA	2023	II, n = 32	Neoadjuvant nivolumab + ipilimumab, MSI-high	pCR 58.6%	Hypothesis-generating
KEYNOTE-585	2024	III, n = 804	Pembrolizumab vs placebo with perioperative chemotherapy	Final analysis: pCR 13.4% vs 2.0%; EFS 44.4 vs 25.7 months, HR 0.81; OS 71.8 vs 55.7, HR 0.86	Negative
ATTRACTION-5	2024	III, n = 755	Adjuvant nivolumab + chemotherapy vs chemotherapy, stage III	3-year RFS 68.4% vs 65.3%, HR 0.90	Negative
INFINITY	2025	II, n = 18	Neoadjuvant tremelimumab + durvalumab, MSI-high	pCR 60%	Hypothesis-generating
MATTERHORN	2025	III, n = 948	Durvalumab vs placebo with perioperative FLOT	2-year EFS 67.4% vs 58.5%, HR 0.71; OS HR 0.78; pCR 19.2% vs 7.2%	FDA Nov 2025
DEWI	2026	II, n = 24	Dostarlimab with watch-and-wait, MSI-high	62.5% without surgery at 1 year	ESMO 2026
SURGERY					
JCOG9502	2006	III, n = 167	Left thoracoabdominal vs transhiatal approach, Siewert II–III	5-year OS 37.9% vs 52.3%	Transhiatal approach
JCOG9501	2008	III, n = 523	D2 with vs without para-aortic dissection	5-year OS 70.3% vs 69.2%	No benefit
Dutch D1D2	2010	III, n = 711	D2 vs D1, 15-year follow-up	Gastric-cancer death 37% vs 48%	D2 standard
JCOG0110	2017	III, n = 505	Splenectomy vs preservation in total gastrectomy	5-year OS 75.1% vs 76.4%; more morbidity	Preserve the spleen
KLASS-01	2019	III, n = 1,416	Laparoscopic vs open distal gastrectomy, stage I	5-year OS 94.2% vs 93.3%	Non-inferior
CLASS-01	2019	III, n = 1,056	Laparoscopic vs open D2 distal gastrectomy, locally advanced	3-year DFS 76.5% vs 77.8%	Non-inferior
KLASS-02	2020	III, n = 1,050	Laparoscopic vs open D2 distal gastrectomy, locally advanced	3-year RFS 80.3% vs 81.3%	Non-inferior
KLASS-05	2023	RCT, n = 138	Proximal gastrectomy with double tract vs total gastrectomy, upper-third cT1N0	Vitamin B12 given 0.4 vs 2.5 mg; 2-year OS 98.5% vs 100%	Function-preserving option

READING THIS PAGE | Systemic therapy around a D2 gastrectomy is the constant. **Radiotherapy added nothing after D2**; durvalumab is the first immunotherapy to add survival.

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C Advanced adenocarcinoma, first line: HER2-negative

Trial	Year	Design	Population and arms	Key result	Status
CHEMOTHERAPY BACKBONE					
V325	2006	III, n = 445	Docetaxel + cisplatin–5-FU vs cisplatin–5-FU	OS 9.2 vs 8.6 months; more toxicity	Triplet not routine
REAL-2	2008	III, n = 1,002	Capecitabine for 5-FU, oxaliplatin for cisplatin	Non-inferior; EOX 11.2 months	Oxaliplatin and capecitabine accepted
GO2	2021	III, n = 514	Oxaliplatin–capecitabine at 100%, 80% or 60% of the dose, frail or older	60% dose non-inferior for PFS, less toxicity	Reduced dose for the frail
GASTFOX	2025	III, n = 507	Docetaxel + FOLFOX vs FOLFOX	PFS 7.6 vs 6.0 months; OS 15.1 vs 12.7	Option for the fit I, B
CHECKPOINT INHIBITORS					
KEYNOTE-062	2020	III, n = 763	Pembrolizumab ± chemotherapy vs chemotherapy, CPS ≥1	OS 12.5 vs 11.1 months with the combination, not significant	Negative
CheckMate 649	2021	III, n = 1,581	Nivolumab + chemotherapy vs chemotherapy	CPS ≥5: OS 14.4 vs 11.1 months, HR 0.71; all patients: 13.8 vs 11.6, HR 0.80	FDA Apr 2021
JAVELIN Gastric 100	2021	III, n = 499	Avelumab maintenance vs continued chemotherapy	OS 10.4 vs 10.9 months	Negative
ATTRACTION-4	2022	III, n = 724	Nivolumab + chemotherapy vs chemotherapy, Asia	PFS 10.5 vs 8.3 months; OS 17.5 vs 17.2, not significant	Regional
CheckMate 649, second comparison	2022	Same trial	Nivolumab + ipilimumab vs chemotherapy, CPS ≥5	OS 11.2 vs 11.6 months	Negative
KEYNOTE-859	2023	III, n = 1,579	Pembrolizumab + chemotherapy vs chemotherapy	OS 12.9 vs 11.5 months, HR 0.78; CPS ≥10: 15.7 vs 11.8	FDA Nov 2023
ORIENT-16	2023	III, n = 650	Sintilimab + CAPOX vs CAPOX, China	OS 15.2 vs 12.3 months, HR 0.77	Regional
RATIONALE-305	2024	III, n = 997	Tislelizumab + chemotherapy vs chemotherapy	OS 15.0 vs 12.9 months, HR 0.80; TAP ≥5%: 17.2 vs 12.6	FDA Dec 2024
FDA pooled analysis	2024	Three trials	PD-1 inhibitor + chemotherapy when PD-L1 is below 1	OS HR 0.91 (0.75–1.09)	Labels restricted, 2025
ATTRACTION-6	2026	III, n = 626	Nivolumab + ipilimumab + chemotherapy vs chemotherapy	OS 15.7 vs 15.8 months, HR 0.90	ASCO 2026 Negative
STAR-221	2026	III	Domvanalimab + zimberelimab + chemotherapy vs nivolumab + chemotherapy	No improvement in survival	ESMO GI 2026 Negative
CLDN18.2 AND FGFR2B					
FIGHT	2022	II, n = 155	Bemarituzumab + mFOLFOX6 vs placebo, FGFR2b-positive	PFS 9.5 vs 7.4 months	Led to phase III
SPOTLIGHT	2023	III, n = 565	Zolbetuximab + mFOLFOX6 vs placebo	PFS 10.6 vs 8.7 months; OS 18.2 vs 15.5, HR 0.75	FDA Oct 2024
GLOW	2023	III, n = 507	Zolbetuximab + CAPOX vs placebo	PFS 8.2 vs 6.8 months; OS 14.4 vs 12.2, HR 0.77	FDA Oct 2024
FORTITUDE-101	2025	III	Bemarituzumab + mFOLFOX6 vs placebo, FGFR2b-positive	OS HR 0.61 early, 0.82 with longer follow-up	Program stopped

READING THIS PAGE | A PD-1 inhibitor helps when **PD-L1 is 1 or more**, zolbetuximab when **CLDN18.2 is positive**. Adding a second checkpoint inhibitor has failed three times.

FDA Aug 2026 US approval, month and year Negative primary endpoint not met Regional data or approval limited to one region ASCO 2026 ESMO 2026 presented in 2026 Year: main publication or presentation. OS overall survival; PFS progression-free survival; DFS disease-free survival; EFS event-free survival; RFS relapse-free survival; pCR pathological complete response; HR hazard ratio.

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C Advanced adenocarcinoma: HER2-positive & later lines

Trial	Year	Design	Population and arms	Key result	Status
HER2-POSITIVE, FIRST LINE					
ToGA	2010	III, n = 594	Trastuzumab + chemotherapy vs chemotherapy	OS 13.8 vs 11.1 months, HR 0.74	FDA Oct 2010
LOGiC	2016	III, n = 545	Lapatinib + CAPOX vs CAPOX	OS 12.2 vs 10.5 months	Negative
JACOB	2018	III, n = 780	Pertuzumab added to trastuzumab + chemotherapy	OS 17.5 vs 14.2 months, HR 0.84	Negative
KEYNOTE-811	2023	III, n = 698	Pembrolizumab vs placebo with trastuzumab + chemotherapy	OS 20.0 vs 16.8 months, HR 0.80; PFS 10.0 vs 8.1	FDA May 2021 CPS ≥1 from Nov 2023
HERIZON-GEA-01	2026	III, n = 914	Zanidatamab + chemotherapy ± tislelizumab vs trastuzumab + chemotherapy	PFS 12.4 vs 8.1 months, HR 0.63–0.65; OS at 35 months 27.0 (triplet) and 22.9 (doublet) vs 19.6	FDA Aug 2026
HER2-POSITIVE, LATER LINES					
GATSBY	2017	II/III, n = 415	Trastuzumab emtansine vs taxane	OS 7.9 vs 8.6 months	Negative
DESTINY-Gastric01	2020	II, n = 187	T-DXd vs chemotherapy, third line, Asia	Response 51% vs 14%; OS 12.5 vs 8.4 months, HR 0.59	FDA Jan 2021
T-ACT	2020	II, n = 91	Paclitaxel ± trastuzumab beyond progression	PFS 3.7 vs 3.2 months	Negative
DESTINY-Gastric04	2025	III, n = 494	T-DXd vs ramucirumab + paclitaxel, second line	OS 14.7 vs 11.4 months, HR 0.70	Second-line standard
KC-WISE	2026	III, n = 246	Anbenitamab + chemotherapy vs chemotherapy, after trastuzumab, China	OS 19.6 vs 9.6 months, HR 0.39	ESMO 2026 Regional
SECOND AND THIRD LINE, ANY HER2 STATUS					
WJOG 4007	2013	III, n = 223	Paclitaxel vs irinotecan	OS 9.5 vs 8.4 months	Either is valid
REGARD	2014	III, n = 355	Ramucirumab vs placebo	OS 5.2 vs 3.8 months, HR 0.78	FDA Apr 2014
RAINBOW	2014	III, n = 665	Ramucirumab + paclitaxel vs paclitaxel	OS 9.6 vs 7.4 months, HR 0.81	FDA Nov 2014
COUGAR-02	2014	III, n = 168	Docetaxel vs active symptom control	OS 5.2 vs 3.6 months, HR 0.67	Chemotherapy helps
ATTRACTION-2	2017	III, n = 493	Nivolumab vs placebo, third line, Asia	OS 5.3 vs 4.1 months, HR 0.63	Regional
KEYNOTE-061	2018	III, n = 592	Pembrolizumab vs paclitaxel, CPS ≥1	OS 9.1 vs 8.3 months, HR 0.82	Negative
TAGS	2018	III, n = 507	Trifluridine–tipiracil vs placebo, third line	OS 5.7 vs 3.6 months, HR 0.69	FDA Feb 2019
RAMIRIS	2026	III	FOLFIRI + ramucirumab vs paclitaxel + ramucirumab, after a taxane	OS 8.7 vs 7.5 months, HR 0.95; PFS 5.5 vs 3.7	ESMO 2026
CLDN18.2 ANTIBODY–DRUG CONJUGATES AND MSI-HIGH DISEASE					
G-HOPE-001	2026	III	Arcotatug tavatecan vs chemotherapy, after 2–4 regimens	OS 10.2 vs 6.9 months, HR 0.57; response 37.4% vs 7.7%	ESMO 2026
CLARITY-Gastric01	2026	III	Sonesitatug vedotin vs chemotherapy	Third line or later: OS 10.0 vs 5.0 months, HR 0.47	ESMO 2026
KEYNOTE-062, MSI-high	2020	Subgroup	Pembrolizumab vs chemotherapy	OS HR 0.29	Supports immunotherapy
CheckMate 649, MSI-high	2022	Subgroup	Nivolumab + chemotherapy vs chemotherapy	OS 38.7 vs 12.3 months, HR 0.38	Supports immunotherapy

READING THIS PAGE | HER2 remains the most rewarding target: **zanidatamab first, trastuzumab deruxtecan second.** In later lines every active drug adds 2–3 months, so fitness decides.

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Advanced esophageal squamous cell carcinoma

Trial	Year	Design	Population and arms	Key result	Status
FIRST LINE					
KEYNOTE-590	2021	III, n = 749	Pembrolizumab + cisplatin–5-FU vs chemotherapy; 73% squamous	OS 12.4 vs 9.8 months, HR 0.73; squamous CPS ≥10: 13.9 vs 8.8, HR 0.57	FDA Mar 2021
ESCORT-1st	2021	III, n = 596	Camrelizumab + paclitaxel–cisplatin vs chemotherapy	OS 15.3 vs 12.0 months, HR 0.70	Regional
CheckMate 648	2022	III, n = 970	Nivolumab + chemotherapy, or nivolumab + ipilimumab, vs chemotherapy	PD-L1 ≥1%: OS 15.4 and 13.7 vs 9.1 months	FDA May 2022
JUPITER-06	2022	III, n = 514	Toripalimab + paclitaxel–cisplatin vs chemotherapy	OS 17.0 vs 11.0 months, HR 0.58	Regional
ORIENT-15	2022	III, n = 659	Sintilimab + chemotherapy vs chemotherapy	OS 16.7 vs 12.5 months, HR 0.63	Regional
RATIONALE-306	2023	III, n = 649	Tislelizumab + chemotherapy vs chemotherapy	OS 17.2 vs 10.6 months, HR 0.66	FDA Mar 2025
ASTRUM-007	2023	III, n = 551	Serplulimab + cisplatin–5-FU vs chemotherapy, CPS ≥1	OS 15.3 vs 11.8 months, HR 0.68	Regional
GEMSTONE-304	2024	III, n = 540	Sugemalimab + chemotherapy vs chemotherapy	OS 15.3 vs 11.5 months, HR 0.70	Regional
SKYSCRAPER-08	2024	III, n = 461	Tiragolumab + atezolizumab + chemotherapy vs chemotherapy, Asia	OS 15.7 vs 11.1 months, HR 0.70	Control had no PD-1 inhibitor
FDA pooled analysis	2024	Three trials	PD-1 inhibitor + chemotherapy when PD-L1 is below 1	OS HR 1.10 (0.76–1.58)	Labels restricted, 2025
SECOND LINE AND BEYOND					
ATTRACTION-3	2019	III, n = 419	Nivolumab vs taxane	OS 10.9 vs 8.4 months, HR 0.77	FDA Jun 2020
KEYNOTE-181	2020	III, n = 628	Pembrolizumab vs chemotherapy	CPS ≥10: OS 9.3 vs 6.7 months, HR 0.69	FDA Jul 2019
ESCORT	2020	III, n = 457	Camrelizumab vs chemotherapy	OS 8.3 vs 6.2 months, HR 0.71	Regional
RATIONALE-302	2022	III, n = 512	Tislelizumab vs chemotherapy	OS 8.6 vs 6.3 months, HR 0.70	FDA Mar 2024
PANKU-Esophagus01	2026	III, n = 497	Izalontamab brengitecan vs chemotherapy after platinum and PD-1 therapy	OS 9.8 vs 7.2 months, HR 0.63; PFS 4.2 vs 2.0, HR 0.50	ASCO 2026 Regional
LOCAL THERAPY AND PALLIATION					
SIREC	2004	III, n = 209	Brachytherapy 12 Gy vs stent for dysphagia	Stent faster; brachytherapy more durable	Both valid
ROCS	2021	III, n = 220	Stent ± palliative radiotherapy	No gain in swallowing or survival; less bleeding	Negative
ESO-Shanghai 13	2024	II, n = 104	Systemic therapy ± local therapy, oligometastatic	PFS 15.3 vs 6.4 months, HR 0.26	Negotiable

READING THIS PAGE | Eight first-line trials agree: a PD-1 or PD-L1 inhibitor with chemotherapy prolongs survival **when PD-L1 is positive.** After immunotherapy, the first new option in years is an antibody–drug conjugate.

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Metastatic surgery, HIPEC & the FDA approval timeline

ESOPHAGUS STOMACH

Trial	Year	Design	Population and arms	Key result	Status
SURGERY AND INTRAPERITONEAL THERAPY IN METASTATIC DISEASE					
REGATTA	2016	III, n = 175	Gastrectomy + chemotherapy vs chemotherapy, one non-curable factor	2-year OS 25.1% vs 31.7%	Negative
AIO-FLOT3	2017	II, n = 252	FLOT, then surgery for limited metastatic disease	OS 31.3 months after resection vs 15.9 without	Hypothesis only
PHOENIX-GC	2018	III, n = 183	Intraperitoneal + intravenous paclitaxel + S-1 vs cisplatin + S-1	OS 17.7 vs 15.2 months	Negative
GASTRIPEC-I	2024	III, n = 105	Cytoreduction ± HIPEC	OS 14.9 vs 14.9 months; PFS 7.1 vs 3.5	Negative
RENAISSANCE	2024	III, n = 139	FLOT ± surgery, limited metastatic disease	No survival advantage	Negative
PERISCOPE II	2026	III, n = 102	Gastrectomy + cytoreduction + HIPEC vs systemic therapy	OS 15.7 vs 16.6 months, HR 1.10; serious adverse events 44% vs 6%	Negative

FDA APPROVALS IN ESOPHAGEAL AND GASTRIC CANCER, 2010–2026

Approved	Drug	Indication in upper-gastrointestinal cancer	Pivotal trial
Oct 2010	Trastuzumab + chemotherapy	HER2-positive metastatic gastric or junctional adenocarcinoma, first line	ToGA
Apr 2014	Ramucirumab	Gastric or junctional adenocarcinoma after first-line chemotherapy	REGARD
Nov 2014	Ramucirumab + paclitaxel	Second line	RAINBOW
May 2017	Pembrolizumab	MSI-high or mismatch-repair deficient solid tumors, any site	Pooled KEYNOTE trials
Feb 2019	Trifluridine–tipiracil	Gastric or junctional adenocarcinoma, third line	TAGS
Jul 2019	Pembrolizumab	Squamous esophageal cancer, second line, CPS 10 or more	KEYNOTE-181
Jun 2020	Nivolumab	Squamous esophageal cancer, second line	ATTRACTION-3
Jan 2021	Trastuzumab deruxtecan	HER2-positive gastric or junctional adenocarcinoma after trastuzumab	DESTINY-Gastric01
Mar 2021	Pembrolizumab + chemotherapy	Esophageal or junctional carcinoma, first line	KEYNOTE-590
Apr 2021	Nivolumab + chemotherapy	Gastric, junctional and esophageal adenocarcinoma, first line	CheckMate 649
May 2021	Pembrolizumab + trastuzumab + chemotherapy	HER2-positive, first line: accelerated; CPS 1 or more from Nov 2023; full approval Mar 2025	KEYNOTE-811
May 2021	Nivolumab, adjuvant	Residual disease after chemoradiotherapy and resection	CheckMate 577
May 2022	Nivolumab + chemotherapy; nivolumab + ipilimumab	Squamous esophageal cancer, first line	CheckMate 648
Nov 2023	Pembrolizumab + chemotherapy	HER2-negative gastric or junctional adenocarcinoma, first line	KEYNOTE-859
Mar 2024	Tislelizumab	Squamous esophageal cancer, second line	RATIONALE-302
Oct 2024	Zolbetuximab + chemotherapy	CLDN18.2-positive, HER2-negative adenocarcinoma, first line	SPOTLIGHT, GLOW
Dec 2024	Tislelizumab + chemotherapy	Gastric or junctional adenocarcinoma, first line, PD-L1 1 or more	RATIONALE-305
Mar 2025	Tislelizumab + chemotherapy	Squamous esophageal cancer, first line, PD-L1 1 or more	RATIONALE-306
May–Jun 2025	Nivolumab and pembrolizumab	First-line labels restricted to PD-L1 of 1 or more	FDA pooled analysis
Nov 2025	Durvalumab + FLOT	Resectable gastric or junctional adenocarcinoma, perioperative	MATTERHORN
Aug 2026	Zanidatamab + chemotherapy, with or without tislelizumab	HER2-positive gastroesophageal adenocarcinoma, first line	HERIZON-GEA-01

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Rx

QUICK REFERENCE

Regimen quick reference

ESOPHAGUS STOMACH

The doses used in the pivotal trials, for quick orientation at the tumor board.

Regimen	Drugs and doses	Schedule	Note
PERIOPERATIVE AND NEOADJUVANT			
FLOT	Docetaxel 50, oxaliplatin 85, leucovorin 200, 5-FU 2,600 mg/m² over 24 hours	Every 2 weeks: 4 cycles before and 4 after surgery	Consider G-CSF
D-FLOT	FLOT + durvalumab 1,500 mg every 4 weeks	2 doses before and 2 after surgery, then every 4 weeks ×10	MATTERHORN
CROSS	Carboplatin AUC 2 + paclitaxel 50 mg/m²	Weekly ×5 with 41.4 Gy in 23 fractions	Surgery 4–8 weeks later
DCF	Docetaxel 70 and cisplatin 70 mg/m² day 1; 5-FU 750 mg/m² daily, days 1–5	Every 3 weeks ×3	JCOG1109; primary G-CSF
Adjuvant nivolumab	240 mg every 2 weeks for 16 weeks, then 480 mg every 4 weeks	1 year in total	CheckMate 577
Adjuvant CAPOX	Oxaliplatin 130 mg/m² day 1; capecitabine 1,000 mg/m² twice daily, days 1–14	Every 3 weeks for 6 months; 4 months non-inferior	CLASSIC, LOMAC
Adjuvant S-1	40–60 mg twice daily by body surface area, days 1–28	Every 6 weeks for 1 year	ACTS-GC
WITH DEFINITIVE RADIOTHERAPY, 50–50.4 GY			
FOLFOX	Oxaliplatin 85, leucovorin 200, 5-FU bolus 400, then 1,600 mg/m² over 46 hours	Every 2 weeks ×6, three of them with radiotherapy	PRODIGE 5
Carboplatin–paclitaxel	Carboplatin AUC 2 + paclitaxel 50 mg/m²	Weekly during radiotherapy	
Cisplatin–5-FU	Cisplatin 75 mg/m² day 1; 5-FU 1,000 mg/m² daily, days 1–4	Weeks 1, 5, 8 and 11	RTOG 85-01
Capecitabine	825 mg/m² twice daily on radiotherapy days	Throughout radiotherapy	Gastric bed, 45 Gy
CHEMOTHERAPY BACKBONES FOR ADVANCED DISEASE			
mFOLFOX6	Oxaliplatin 85, leucovorin 400, 5-FU bolus 400, then 2,400 mg/m² over 46 hours	Every 2 weeks	No bolus with zanidatamab
CAPOX	Oxaliplatin 130 mg/m² day 1; capecitabine 1,000 mg/m² twice daily, days 1–14	Every 3 weeks	
Cisplatin–5-FU	Cisplatin 80 mg/m² day 1; 5-FU 800 mg/m² daily, days 1–5	Every 3 weeks; cisplatin up to 6 cycles	KEYNOTE-590
FOLFIRI	Irinotecan 180, leucovorin 400, 5-FU bolus 400, then 2,400 mg/m² over 46 hours	Every 2 weeks	
ANTIBODIES AND IMMUNOTHERAPY			
Nivolumab	240 mg every 2 weeks or 360 mg every 3 weeks, with chemotherapy	To progression, or 2 years	
Pembrolizumab	200 mg every 3 weeks or 400 mg every 6 weeks	To progression, or 2 years	
Tislelizumab	200 mg every 3 weeks	To progression	
Nivolumab + ipilimumab	Nivolumab 3 mg/kg every 2 weeks + ipilimumab 1 mg/kg every 6 weeks	To progression, or 2 years	Squamous cancer
Trastuzumab	8 mg/kg, then 6 mg/kg	Every 3 weeks	Ejection fraction every 3 months
Zanidatamab	1,800 mg under 70 kg; 2,400 mg at 70 kg or more	Every 3 weeks	Diarrhea prophylaxis in cycle 1
Zolbetuximab	800 mg/m² first dose; then 600 mg/m² every 3 weeks or 400 mg/m² every 2 weeks	With the chemotherapy cycle	Antiemetics; slow first infusion
Dostarlimab	500 mg every 3 weeks ×4, then 1,000 mg every 6 weeks	MSI-high disease	
LATER LINES			
Ramucirumab + paclitaxel	Ramucirumab 8 mg/kg days 1 and 15; paclitaxel 80 mg/m² days 1, 8 and 15	Every 4 weeks	RAINBOW
Trastuzumab deruxtecan	6.4 mg/kg	Every 3 weeks	Lung surveillance
Trifluridine–tipiracil	35 mg/m² twice daily, days 1–5 and 8–12	Every 4 weeks	TAGS
Irinotecan	150–180 mg/m²	Every 2 weeks	
Docetaxel	75 mg/m²	Every 3 weeks	
Paclitaxel	80 mg/m² days 1, 8 and 15	Every 4 weeks	

BEFORE PRESCRIBING

- **These are trial doses, not prescriptions.** Check the current label, organ function, interactions and your local protocol before every order.
- ***DPYD* genotype** before the first fluoropyrimidine (EMA 2020; FDA boxed warning 2025–26); reduce the dose in carriers of a reduced-function variant, and do not treat complete deficiency.
- **Renal function and hearing** before cisplatin; **neuropathy** before every oxaliplatin or taxane dose.

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Edition 1.1 · 13 November 2026: what was re-checked in this revision

- **Every published trial** in the compendium and the chapters was compared with its PubMed record: sample size, medians, hazard ratios and rates. The reference list grew from 91 to 128 entries, with a DOI for every published paper.
- **Updated to the full publications of 2026:** HERIZON-GEA-01 (N Engl J Med), MATTERHORN final survival (Lancet), PANKU-Esophagus01 (JAMA), SKYSCRAPER-07 and KC-WISE (Ann Oncol), PERISCOPE II and SKYSCRAPER-08 (Lancet Oncol); *DPYD* testing as now recommended in the FDA boxed warning.
- **Corrected:** frequency of MSI-high disease by site (about 8% of resectable gastric cancers, under 1% in the esophagus); radiotherapy dose limits aligned with NCCN; hazard ratio of PANKU-Esophagus01 (0.63); sample sizes of RTOG 85-01 and RESOLVE; designs of JCOG0508 and JCOG0502; sources of the second-primary, hereditary-testing and early-disease statements.

Edition 1.2 · 13 November 2026: the 3D illustrated edition

- **Thirteen three-dimensional plates** replace the line drawings of the anatomy, endoscopic dissection, esophagectomy, gastrectomy and lymph-node map (chapters 3, 4 and 8); two plate pages were added. The schematic drawings that carry measurements (Siewert types, T category, depth of invasion, radiotherapy plans) are kept.
- **Text:** every correction of edition 1.1 is retained. New on the plate pages: the AJCC definition of T4b in the stomach, the JGCA station of each artery, and two randomized sources on reconstruction (KLASS-05 and a Cochrane review), both checked against PubMed. The list now holds 130 references.

A note on accuracy. Figures are quoted from the publications and presentations listed here. Congress data of 2026 that are not yet published in full may change when they are. If you find an error, please tell the author. Medicine moves: confirm against the current guideline version and the product label before applying anything here to a patient.

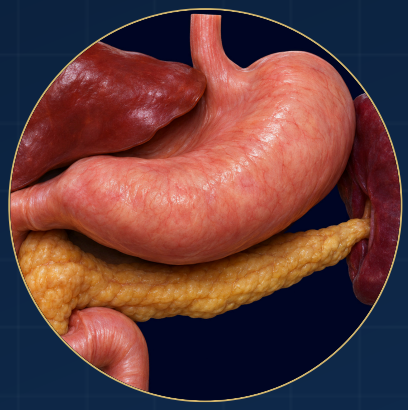
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UPPER GASTROINTESTINAL ONCOLOGY

Gastric & Esophageal Cancer Management 2026

From Diagnosis & Staging to Survivorship



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13
three-dimensional plates of anatomy and surgery

108
trials in the compendium, with FDA approval dates

32
non-negotiable steps on one checklist page

130
references, current to 13 November 2026

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